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CNN based skin lesion classification

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Abstract

This project focused on classifying seven skin lesion types based on dermoscopic images using deep learning techniques. It was carried out in two main stages: one using the unbalanced dataset originally used for experiments, and the other after applying a hybrid resampling and data augmentation approach to tackle severe class imbalance. In the first stage, a customized CNN, alongside a pre-trained EfficientNet-B0 architecture, was employed and benchmarked against each other. During the second phase, four models consisting of Customized CNN, EfficientNet-B0, ResNet50, and VGG16 were experimented on using a dataset of 1,000 images per class for each class. The metrics used were accuracy, macro-F1, weighted-F1, and macro-ROC-AUC scores. Experiment results revealed that balancing the dataset and transfer learning contributed significantly to minority class detection. ResNet50 provided the best overall performance by achieving an accuracy of 86.3%, a macro F1 score of 0.861, and a macro-ROC-AUC score of 0.9805. The outcomes validate the potential of deep learning when integrated with proper dataset preparation and model design in clinical decision-making systems.

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Glossary

The following terms, acronyms, and abbreviations are used throughout this report.

Term / Acronym	Definition
AKIEC	Actinic Keratoses and Intraepithelial Carcinoma — a pre-malignant skin lesion class in HAM10000.
AUC	Area Under the Curve — the area beneath an ROC curve; higher values indicate better discrimination ability.
BCC	Basal Cell Carcinoma — a common type of skin cancer included in the HAM10000 dataset.
BKL	Benign Keratosis-like lesions — a category of non-cancerous skin growths in HAM10000.
CNN	Convolutional Neural Network — a class of deep learning model designed for image recognition tasks.
DF	Dermatofibroma — a benign fibrous skin nodule; a minority class in HAM10000.
EfficientNet-B0	A compact and efficient CNN architecture that scales depth, width, and resolution simultaneously.
F1-Score	The harmonic mean of precision and recall; used to evaluate classification performance, especially for imbalanced datasets.
GPU	Graphics Processing Unit — hardware used to accelerate deep learning training.
HAM10000	Human Against Machine with 10,000 training images — a publicly available dermoscopic image dataset of 10,015 images across 7 lesion classes.
ImageNet	A large-scale image dataset is used to pre-train deep learning models for transfer learning.
MEL	Melanoma — a malignant and potentially life-threatening skin cancer class in HAM10000.
NV	Melanocytic Nevi — the majority class in HAM10000, representing common moles.
PyTorch	An open-source deep learning framework used for model development in this project.
ResNet50	Residual Network with 50 layers — a deep CNN architecture that uses skip connections to prevent vanishing gradients.
ROC Curve	Receiver Operating Characteristic curve — plots true positive rate against false positive rate across classification thresholds.
ROC-AUC	Area under the ROC curve — used here in a one-vs-rest (OvR) multiclass formulation.

Term / Acronym	Definition
VASC	Vascular Lesions — a minority class in HAM10000 representing vascular skin lesions.
VGG16	Visual Geometry Group network with 16 layers — a deep CNN architecture from the University of Oxford.

1. Introduction

1.1 Background and Motivation

Skin cancer is one of the frequently diagnosed cancers worldwide, early and accurate diagnosis is an essential determinant of patient outcomes. Dermoscopy, or the noninvasive optical examination of skin lesions, is now widely used in clinical practice and has improved diagnostic accuracy beyond that achievable with naked eye examination. However, dermoscopic image explanation is highly variable and inclined to diagnostic error, depending on the skill and experience of the clinician, especially in settings with limited access to specialists.

Deep learning methods, especially convolutional neural networks, have recently displayed advanced performance in medical image analysis tasks, such as classifying skin lesion classifying. The publicly available dataset of 10,015 labeled dermoscopic images for seven lesion categories HAM10000 has facilitated a systematic assessment of such approaches.

Despite this availability, several challenges persist. Chief among these is the severe class imbalance inherent in the dataset, where the melanocytic nevi (NV) class alone accounts for most samples, leading models trained without corrective measures to exhibit considerable bias toward the majority class at the expense of minority-class recognition.

This project was motivated by the need to explore different deep learning architectures comparing skin lesion classification and how dataset balancing techniques interact with transfer learning to Manipulate the overall model behavior. The importance of improving recall for minority classes is quite high. Conditions like melanoma carry a risk of death if not caught. In cases a false negative can cost much more clinically than a false positive.

1.2 Aims and Objectives

The overall objective of the study is to design and evaluate deep learning models that could automate various classes of skin lesions based on dermoscopy images of the HAM10000 data set. The study addresses the impact of multiclass class imbalance and transfer learning.

The project's objectives:

- To implement and evaluate a custom CNN architecture on the original unbalanced HAM10000 dataset as a standard model.
- To apply transfer learning using EfficientNet-B0 on the unbalanced dataset and compare its performance versus the custom CNN model.
- To develop a combination of resampling and data augmentation techniques for dealing with class imbalance within the data set.

- Train and evaluate different models; a modified CNN, EfficientNet-B0, ResNet50, and VGG16 models on the balanced dataset through constant evaluation measures.
- Evaluate models on accuracy, macro F1, weighted F1, and macro-ROC-AUC to find the best model architecture for classification purposes.
- Identify performance patterns for each class, detect issues related to misclassification, and discuss the future possibilities and constraints for this project

1.3 Proposed Methods

This project follows the supervised deep learning approach. All models were implemented in PyTorch with CUDA enabled GPU acceleration. The images were normalized with ImageNet means and standard deviation values and resized to 224×224 pixels. In the unbalanced phase, basic preprocessing was applied without augmentation. In the balanced phase, a hybrid resampling strategy was implemented down sampling majority classes and up sampling minority classes through augmentation techniques such as rotation, flipping, affine transformation, color jitter, and zooming to produce a balanced dataset of 1,000 image per class. Transfer learning models were loaded with pretrained ImageNet weights and selectively fine tuned. Evaluations were done using test data sets through classification reports, confusion matrix, ROC curve analysis, and multiclass ROC-AUC analysis.

1.4 Report Overview

- Chapter 2: A brief literature review of the relevant work, provided with an introduction to skin cancer detection, features of the HAM10000 database, and previous deep learning approaches used for skin lesion classification.
- Chapter 3: addresses the project methodology regarding data set preparation, architecture design, configuration of the model learning process, and testing methods.
- Chapter 4: describes the experimental results of the unbalanced and balanced phases, respectively, with figures and quantitative comparisons.
- Chapter 5: represents a discussion of findings in connection with literature review and identification of factors that influenced model behavior.
- Chapter 6: provides the conclusion regarding the objectives mentioned in the introduction and recommendations for future work.

2. Literature Review

2.1 Skin Cancer and the Role of Dermoscopy

Skin cancer can be described as the biggest health challenge facing people all around the world today. Melanoma cancer, even though it is not so common among other cancers, leads to many more fatalities because of its high potential for metastasis. There exists a correlation between early diagnosis and increased survival rates; therefore, software tools for diagnosing skin cancer were developed by experts. The use of dermoscopy allows for better visualization of structures below the lesion. Consequently, an increasing number of dermatologists use it, hence the need for automated systems that can analyze a great deal of dermoscopic images.

2.2 The HAM10000 Dataset

The HAM10000 data set, proposed in association with the ISIC 2018 challenge, consists of 10,015 dermoscopy images that are classified into seven categories including Nevus (NV), Melanoma (MEL), Benign keratosis-like lesions (BKL), Basal cell carcinoma (BCC), Actinic keratoses & intraepithelial carcinoma (AKIEC), Vascular Lesions (VASC), and Dermatofibroma (DF) (Tschandl, Rosendahl and Kittler, 2018). This data set has been noted for having a problem of class imbalance due to NV being the largest class, whereas DF, VASC, and AKIEC are the minority classes. This issue has consistently been a major point of concern in literature that finds models trained on such imbalanced classes having biases toward majority classes.

2.3 Deep Learning Approaches to Skin Lesion Classification

Several studies have used convolutional neural networks for classification in dermoscopy images, with the HAM10000 dataset being a popular choice for testing these methods. However, the custom CNN model, although efficient in computation and implementation, has been found to perform less accurately and less efficiently in detecting minority classes compared to the transfer learning method. As per Ingle and Shaikh (2024), while the custom CNN model attained an accuracy rate of 66%, the VGG16 transfer learning model achieved an accuracy rate of 88.83% using the HAM10000 dataset. Both the models were found to confuse melanoma with nevi due to their similar appearances. Similarly, Wang et al. (2024) demonstrated that YOLOv8 classification variants, when fine-tuned on HAM10000, achieved accuracies ranging from 84.0% to 86.2%, with YOLOv8x-cls performing best, though minority class recall remained a persistent challenge.

Transfer learning based on pretrained ImageNet models has become the common technique in this field. Nguyen et al. (2025) examined the VGG19, ResNet50, and some variations of the YOLOv11 classification model on the HAM10000 dataset, finding that while VGG19 achieved an accuracy score of 89.68% and ResNet50 reached 87.51% on the augmented balanced data set, on the other hand their scores on the unbalanced data set were 79.12% and 77.43%. This suggests that traditional multi-stage networks perform significantly better than single-stage detection-focused networks such as YOLO, and that balancing the dataset through data augmentation improves overall performance.

In addition, another comprehensive comparison was conducted by Alenezi et al. (2024), where several CNN models were examined, including ConvNeXtXLarge, EfficientNetB0, ResNet50, MobileNetV2, and VGG16 on HAM10000 data set. According to Alenezi et al. (2024), the most effective model is ConvNeXtXLarge with 87.62% accuracy. However, the problem of poor classification of some classes remains, since high accuracy is achieved for the whole sample only, While DF was classified by ConvNeXtXLarge with 41.18%, and MEL class with 59.28%.

A large-scale benchmarking study by Vieira et al. (2025) evaluated 38 CNN architectures across 10 model families on the HAM10000 dataset, using transfer learning throughout. Their findings confirmed that larger and more modern architectures such as ConvNeXt consistently outperform traditional networks such as VGG16 and MobileNetV2, while also highlighting that per-class performance, particularly for minority classes including DF and MEL, remains substantially below overall accuracy figures across all evaluated models.

2.4 Class Imbalance and Data Augmentation

Class imbalance is a natural characteristic of medical imaging datasets. For example, in the HAM10000 dataset, NV accounts for about 67% of the data, while DF and VASC together account for less than 3%. In such a case, the model blindly learns to predict the classes that dominate the most, resulting in poor recall for minority classes; in other words, many errors occur when identifying rare lesion types, which is a mistake in clinical practice.

There are different approaches to this problem in literature. Random oversampling takes the training samples from the minority classes and duplicates them; in contrast, augmentation-based up sampling techniques use geometric and color transformations to create synthetic variations (Shorten and Khoshgoftaar, 2019). Majority-class down sampling reduces the training set size, yet it does not eliminate dominating statistically abundant classes. A hybrid down-sampling approach has been best applied to yield balanced learning dynamics and higher per-class recall in the case of augmentation. The choice of augmentation- rotation, flipping, zoom, color jitter, affine transformation, etc.- depends on the characteristics of the target domain. Since dermoscopic images are rotation invariant, rotational augmentation is well motivated.

2.5 Summary and Research Gap

Literature clearly indicates that transfer learning, balancing of the data set, and proper fine-tuning all play an important role in improving the classification accuracy of the skin lesion problem. The ResNet50 and different versions of EfficientNet have shown promise in several independent studies. However, few papers show a detailed comparison between the unbalanced and balanced training approach on the same architectural setup, thus making it tough to determine the impact of the balancing process. The current project seeks to fill this void by performing a comparative study in two phases: First, experiments will be performed for both CNN and EfficientNet under the unbalanced condition, which will then be compared against balanced training involving four models.

3. Project Work and Methodology

3.1 Overview

This chapter describes the experimental approaches applied through both phases and the complete methodology of the whole project. The work was divided into two phases. Phase 1 used the original unbalanced HAM10000 dataset and trained a custom CNN and EfficientNet-B0. In Phase 2, a hybrid sampling technique was used for the creation of a balanced dataset, and multiple architectures like custom CNN, EfficientNet-B0, ResNet50, and VGG16 were trained. All architectures were implemented using PyTorch and trained on a CUDA-enabled GPU. The details are provided in the following sections.

3.2 Dataset

Dataset link:

<https://www.kaggle.com/datasets/surajhuwalewala/ham1000-segmentation-and-classification>

3.2.1 Source and Composition

The HAM10000 dataset was taken as the main source for data. The HAM10000 dataset includes 10,015 dermoscopic images in seven classes. They are in two different folders with image metadata being provided in a CSV file. A helper function was implemented to locate each image across both directories by constructing the full file path from the image identifier. The label encoder from scikit-learn was applied to convert string class labels to integer indices, producing the mapping: akiec=0, bcc=1, bkl=2, df=3, mel=4, nv=5, vasc=6.

The dataset was examined for data integrity issues. No duplicate image IDs were found. Several lesion IDs were associated with multiple images, and file paths were verified to be unique across both image directories.

3.2.2 Class Distribution — Unbalanced Dataset

Figure 3.1 shows the distribution of classes in the training set for the initial unbalanced dataset. There is a clear dominance of class NV, containing about 4,300 training images, while class DF and class VASC have less than 150 images each. This extreme imbalance directly motivated the balancing strategy chosen in Phase 2.

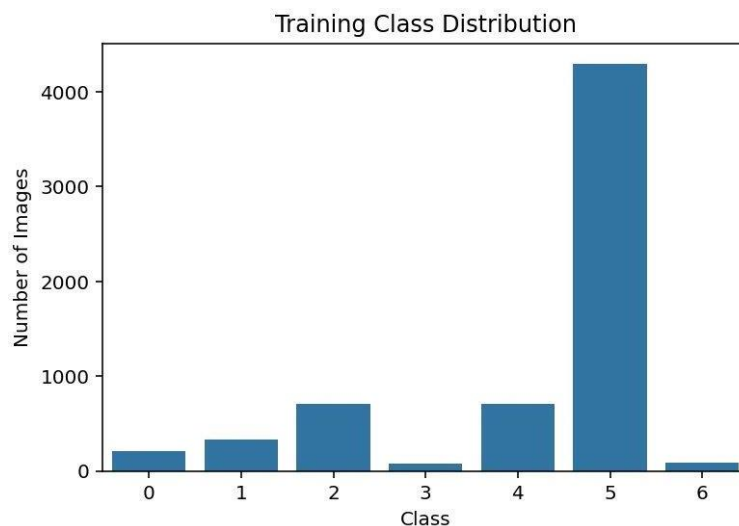


Figure 3.1: Training class distribution in the original unbalanced HAM10000 dataset.

3.2.3 Data Splitting

Phase 1: The entire dataset was divided into training set (80%), validation set (10%), and testing set (10%) using stratified random sampling. Phase 2: The balanced dataset was divided into training set (80%), validation set (10%), and testing set (10%) using stratified random sampling.

3.3 Data Balancing Strategy (Phase 2)

A hybrid resampling strategy was implemented to address the extreme class imbalance identified in the original dataset. A target size of 1,000 samples per class was established. The three majority classes NV, MEL, and BKL were randomly down sampled to 1,000 image each. The four minority classes AKIEC, DF, VASC, and BCC were up sampled to 1,000 image each by data augmentation techniques such as rotation, flipping, affine transformation, color jitter, and zooming.

Figure 3.2 shows representative sample images for each class from the balanced dataset, illustrating the visual diversity across lesion categories. Figure 3.3 demonstrates the data augmentation pipeline applied to training samples in Phase 2.



Figure 3.2: Sample images per class from the balanced dataset after resampling.

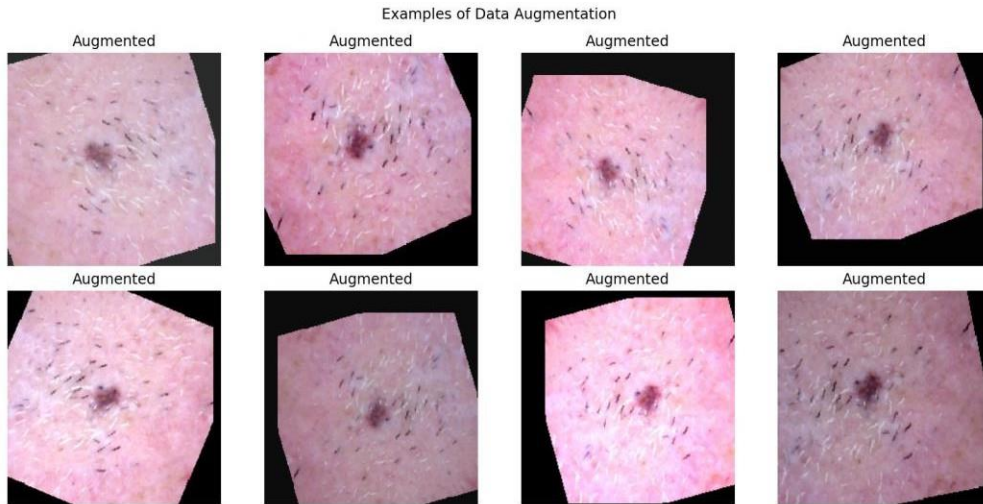


Figure 3.3: Examples of data augmentation applied to a single training image, showing variation in orientation, zoom, and brightness.

Augmentation pipeline for the training phase included: Random horizontal flipping, random vertical flipping, random rotation by 25 degrees, random affine transformation ($\pm 10\%$ translation, 0.9-1.1 scaling), and color jittering (brightness, contrast, and saturation ± 0.25). Validation and test images were processed using only resizing and normalization, without augmentation. Additionally, a separate, less heavy data augmentation strategy was applied for VGG16 because it was more sensitive to pixel level changes (rotation ± 15 degrees, color jitter ± 0.1). The images were resized to 224x224 pixels and standardized based on the mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

3.4 Model Architectures

3.4.1 Phase 1 — Custom CNN (Unbalanced)

The Phase 1 customized CNN consisted of three convolution layers, each containing a convolutional layer (3×3), ReLU activation function, and a max pooling layer (2×2). The filter depths used here were 64, 128, and 256. The network had an adaptive average pooling layer that downsized the image size to 1×1 and then a fully connected classifier consisted of linear layers (256×128 , followed by 128×7), along with ReLU and dropout.

3.4.2 Phase 1 — EfficientNet-B0 (Unbalanced)

EfficientNet-B0 pretrained on ImageNet was loaded with all layers frozen. The final feature extraction block (features [-1]) and the classifier layer were unfrozen to allow specific task adaptation. The original classifier was replaced with a linear layer mapping to the seven output classes. The Adam optimizer was used with a learning rate of 3×10^{-5} , applied only to trainable parameters.

3.4.3 Phase 2 — Custom CNN (Balanced)

The second phase's CNN model was considerably altered compared to its first-phase counterpart. There were five convolutional layers with filter depths of 32, 64, 128, 256, and 512 filters, respectively, followed by batch normalization, rectified linear unit (ReLU) activation, and max-pooling operations. Adaptive average pooling was then performed before the classification model comprised of a linear layer (512 -

>256), ReLU, dropout (0.5), and a seven-class classification layer. The learning rate was dropped from 3×10^{-5} , which was optimized using the Adam optimizer.

3.4.4 Phase 2 — EfficientNet-B0 (Balanced)

The balanced phase EfficientNet-B0 unfroze the last two feature extraction blocks (features [-1] and features [-2]) rather than only the final block used in Phase 1, enabling deeper adaptation to the skin lesion domain. The classifier was replaced with a dropout layer ($p=0.4$) followed by a linear output layer. Training used a learning rate of 1×10^{-4} with a ReduceLROnPlateau scheduler (factor=0.5, patience=3) and proceeded for 50 epochs.

3.4.5 Phase 2 — ResNet50 (Balanced)

The ResNet50 architecture pretrained on ImageNet was loaded where all the parameters were frozen except for the last residual block (layer4) and the classifier. The classification head which consisted of only one layer was changed to a multilayered head consisting of Linear (2048->1024) -> BatchNorm -> ReLU -> Dropout (0.4) -> Linear (1024->512) -> BatchNorm -> ReLU -> Dropout (0.3) -> Linear (512->7). The training was done using the Adam optimizer with the learning rate being 3×10^{-5} and weight decay as 5×10^{-5} along with ReduceLROnPlateau and Early Stopping.

3.4.6 Phase 2 — VGG16 (Balanced)

VGG16 pretrained on ImageNet was loaded with the first 20 convolutional feature layers frozen. The remaining deeper feature layers were left trainable. The full classifier was replaced with: Linear (25088 -> 1024) -> ReLU -> Dropout (0.5) -> Linear (1024->256) -> ReLU -> Dropout (0.4) -> Linear (256->7). Training used the Adam optimizer at a learning rate of 1×10^{-5} with weight decay 1×10^{-4} , a ReduceLROnPlateau scheduler, and early stopping with patience of 5 epochs.

3.5 Training Configuration

All models used cross entropy loss as the objective function. A batch size 64, was used throughout. Training and validation accuracy and loss were recorded at each epoch. Checkpoint systems were implemented for all Phase 2 models to save model weights, optimizer states, and training histories at regular intervals or upon validation loss improvement, enabling training resumption and later evaluation. Table 3.1 summarizes the key training parameters.

Model	Phase	Epochs	Learning Rate	Optimizer	Scheduler
Custom CNN	1 (Unbalanced)	60	1×10^{-4}	Adam	None
EfficientNet-B0	1 (Unbalanced)	60	3×10^{-5}	Adam	None
Custom CNN	2 (Balanced)	100	3×10^{-5}	Adam	None

Model	Phase	Epochs	Learning Rate	Optimizer	Scheduler
EfficientNet-B0	2 (Balanced)	50	1×10^{-4}	Adam	ReduceLROnPlateau
ResNet50	2 (Balanced)	≤ 50 (early stop)	3×10^{-5}	Adam	ReduceLROnPlateau
VGG16	2 (Balanced)	≤ 50 (early stop)	1×10^{-5}	Adam	ReduceLROnPlateau

Table 3.1: Summary of training hyperparameters for all models.

3.6 Evaluation Methodology

The model was evaluated against persevering test datasets based on the following performance measures: accuracy, precision, recall, and F1-score per-class, macro-averaged F1-score, weighted-average F1-score, confusion matrix, class-specific ROC curve, and multiclass macro-ROC-AUC. This multiclass macro-ROC-AUC considers all classes equally regardless of their sizes, making it highly insightful when dealing with data imbalance problems. The SoftMax probability outputs were used to compute the ROC-AUC values. The scikit-learn classification report utility was used to obtain precision, recall, and F1-score values. The final evaluation score for Phase 2 models was calculated as the weighted average: $\text{Accuracy} \times 0.35 + \text{Macro_F1} \times 0.35 + \text{Weighted_F1} \times 0.20 + \text{ROC_AUC} \times 0.10$.

3.7 Alternative Approaches

Various other design options have been considered for the project. The option of weighted cross entropy loss, which penalizes errors in the minority class, was found as an alternative to the hybrid method of up sampling and down sampling, but this option was ruled out at this stage since its effectiveness needs to be studied separately. Combining predictions of different models through ensemble techniques is another technique to enhance accuracy; however, this method requires significant computational power, and hence, it has been left for future research. Generating images using GANs could be seen as an advanced form of up sampling, but it would require a large amount of computation power and training data; thus, this method was not explored within this project's scope.

4. Testing, Results, Discussion and Evaluation

4.1 Phase 1: Unbalanced Dataset Experiments

4.1.1 Custom CNN — Training Behavior

Phase1: 60 epochs were trained for the custom CNN on the unbalanced dataset without data augmentation. Figures 4.1 and 4.2 respectively show the training and validation accuracy and loss curves. Both curves showed slow, steady improvement over the training period, with training accuracy reaching approximately 71.4% and validation accuracy reaching approximately 72.5% by epoch 60. These results of training and validation curves show that the model was not overfitting; however, the overall performance remained low, reflecting the shallow network and difficulty of classification.

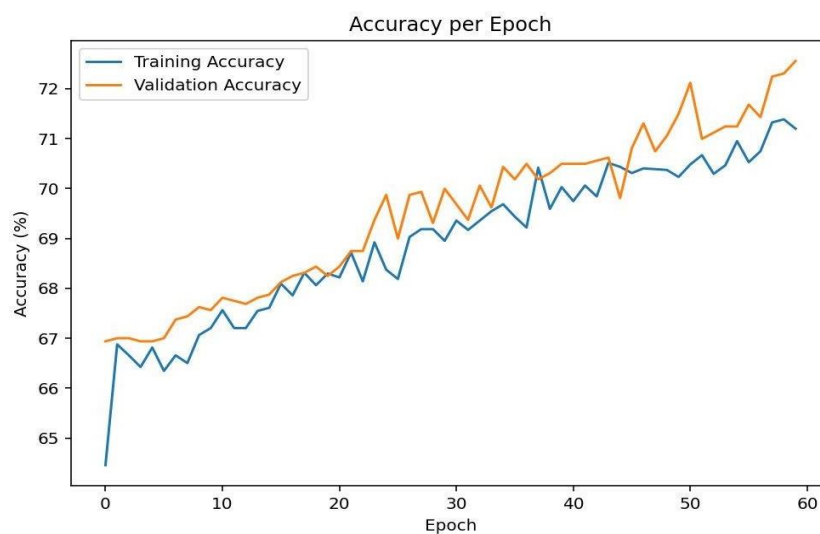


Figure 4.1: CNN (unbalanced) training and validation accuracy per epoch.

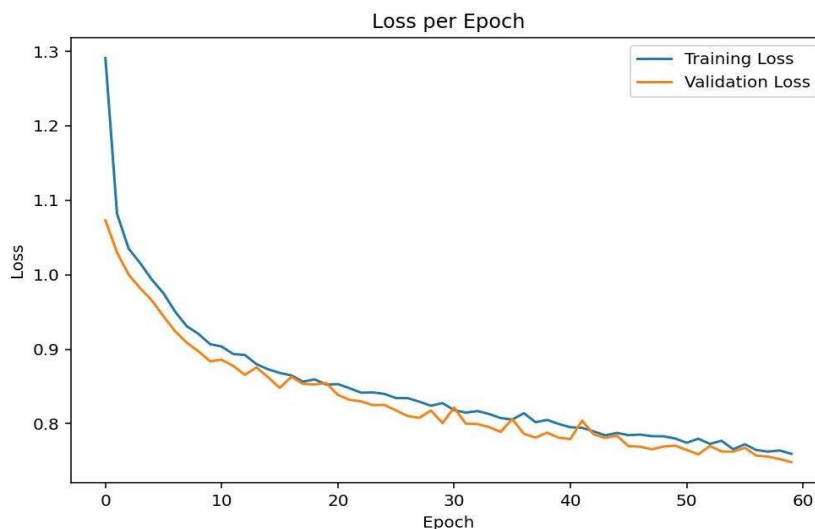


Figure 4.2: CNN (unbalanced) training and validation loss per epoch.

The confusion matrix (Figure 4.3) shows that the model strongly biased toward the NV class, assigning most of the predictions in most classes to NV. Class bias was expected as the data set was imbalanced, leading to low recall for minority classes.

Sample predictions (Figure 4.4) confirm that the model correctly predicts several NV cases and misclassified BKL and AKIEC images.

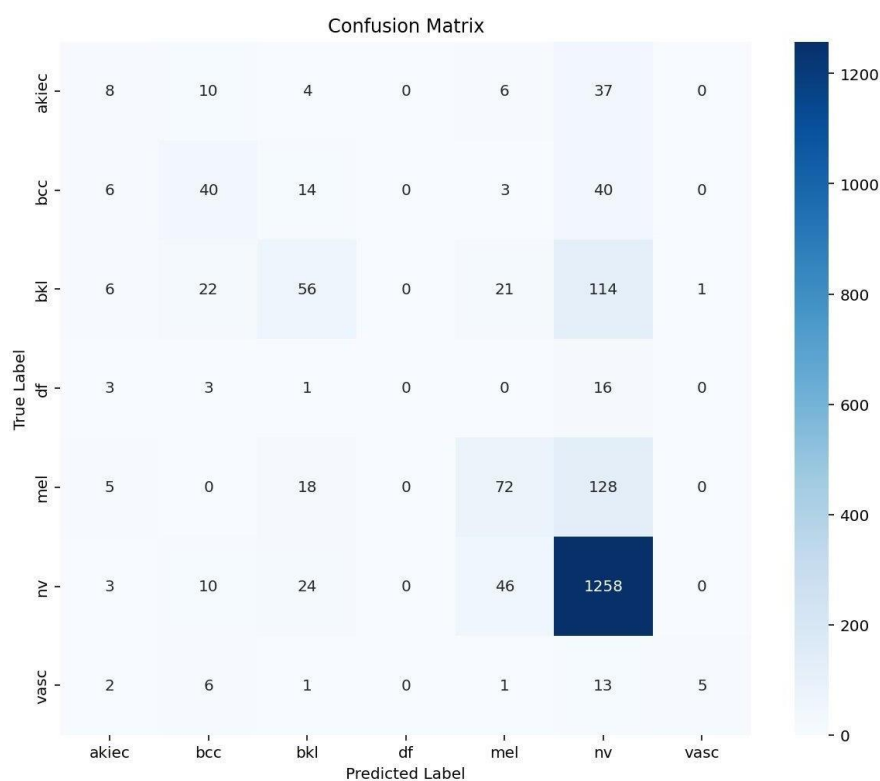


Figure 4.3: Confusion matrix for the CNN trained on the unbalanced dataset.

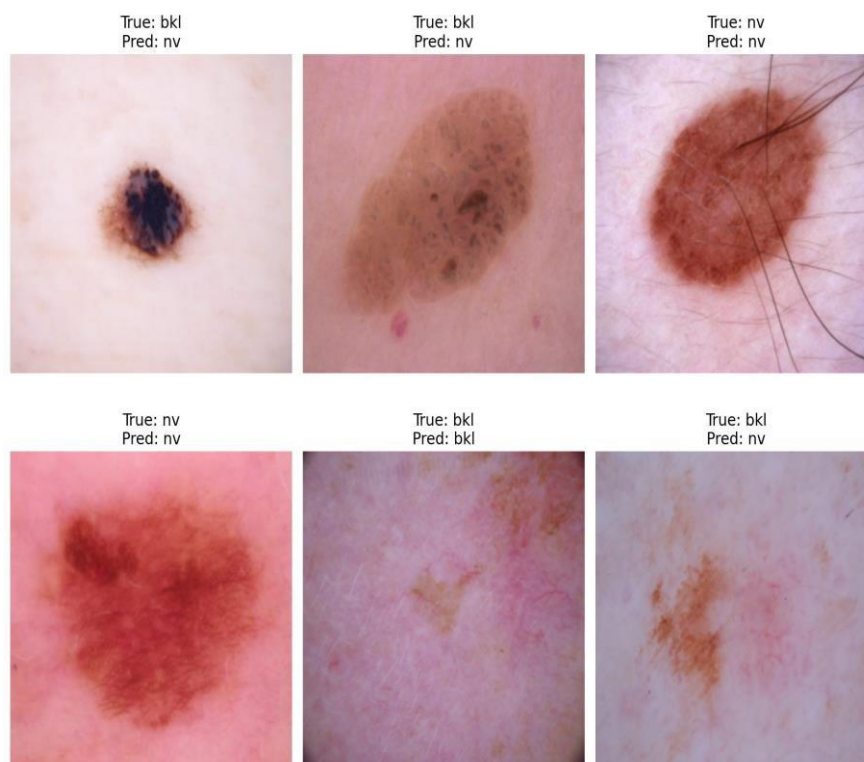


Figure 4.4: Sample predictions from the CNN (unbalanced). Titles show true and predicted class labels.

4.1.2 EfficientNet-B0 — Training Behavior (Unbalanced)

EfficientNet-B0 with transfer learning achieved significantly stronger results for the unbalanced dataset. Training accuracy was approximately 84.7% and validation accuracy approximately 79.6% after 60 epochs (Figure 4.5). The training loss converged smoothly from approximately 1.68 to 0.45, while validation loss stabilized around 0.58 (Figure 4.6). The difference between training and validation can be viewed as a sign of overfitting, which is not surprising since no regularization was introduced in Phase 1.

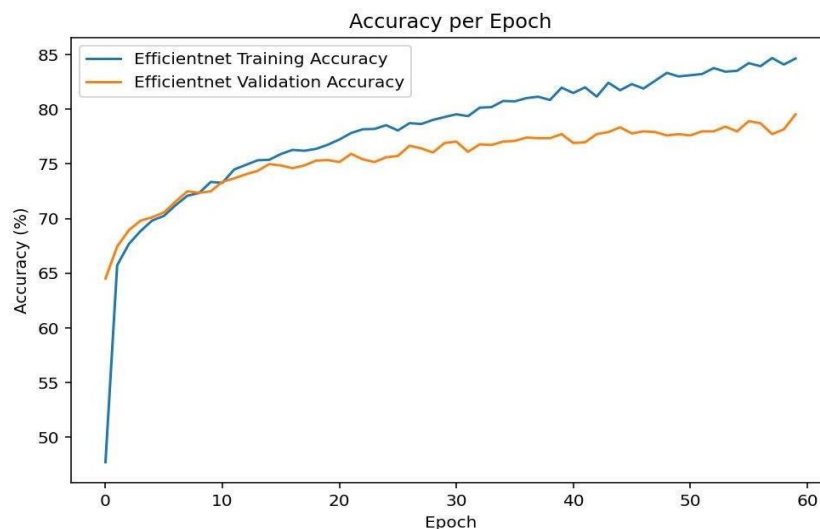


Figure 4.5: EfficientNet-B0 (unbalanced) training and validation accuracy per epoch.

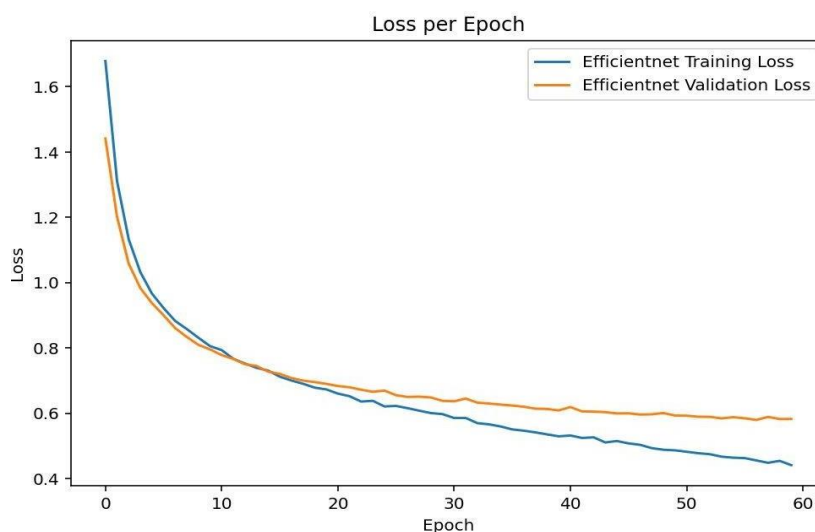


Figure 4.6: EfficientNet-B0 (unbalanced) training and validation loss per epoch.

The confusion matrix shown in Figure 4.7 shows that the classifier is outperforming the CNN in terms of having slightly more dominant diagonal values for some classes. However, NV continued to be dominant in the predictions, and minority class recall was still low. The test predictions shown in Figure 4.8 confirm that EfficientNet is still biased toward NV as it only predicts it.

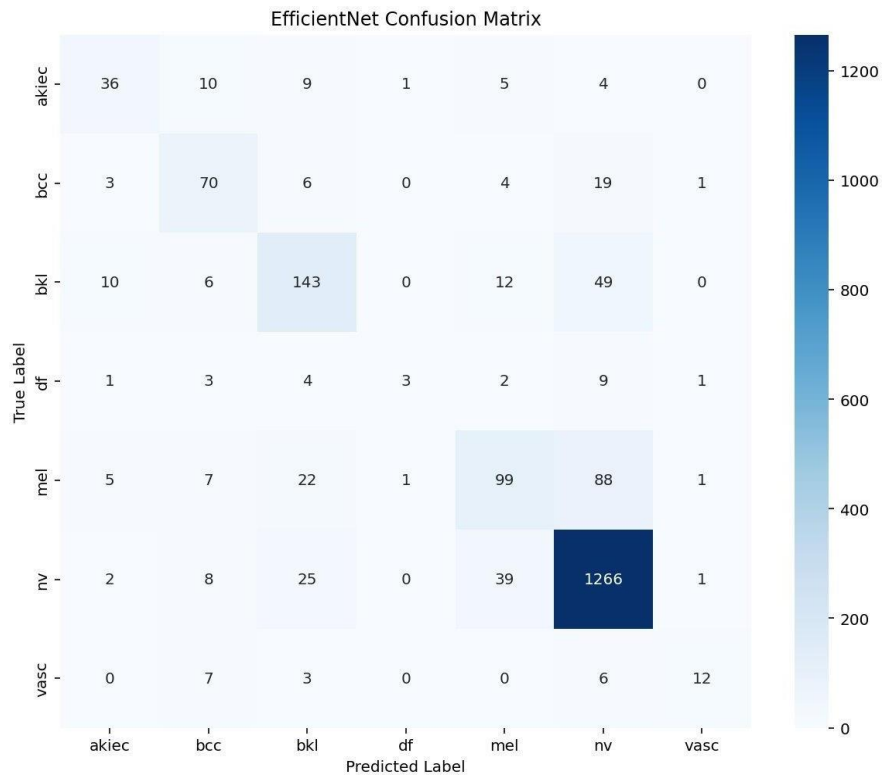


Figure 4.7: Confusion matrix for EfficientNet-B0 trained on the unbalanced dataset.

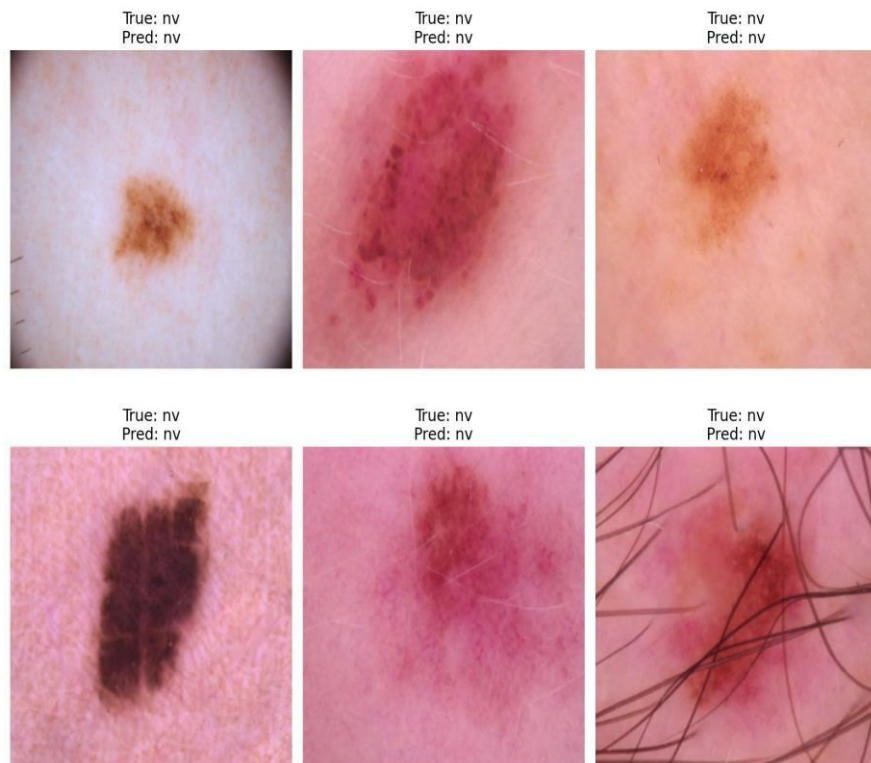


Figure 4.8: Sample predictions from EfficientNet-B0 (unbalanced).

4.1.3 Phase 1 Comparative Analysis

Figure 4.9 shows the F1-score of each class of the two Phase 1 models. EfficientNet performed better than CNN in all seven classes, especially in the NV and BKL

classes. In contrast, the DF class was the hardest class for EfficientNet, where it achieved an F1-score of 0.213, owing to the few DF data in the dataset.

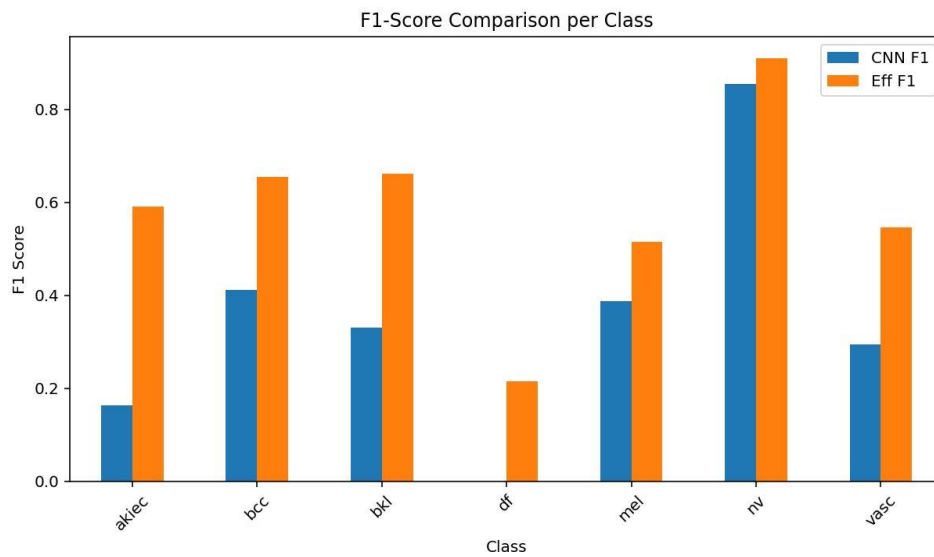


Figure 4.9: Per-class F1-score comparison between CNN and EfficientNet-B0 (unbalanced dataset).

Figure 4.10 presents the overall performance comparison, and Figure 4.11 the ROC-AUC evaluation summary. Table 4.1 summaries the final comparison metrics.

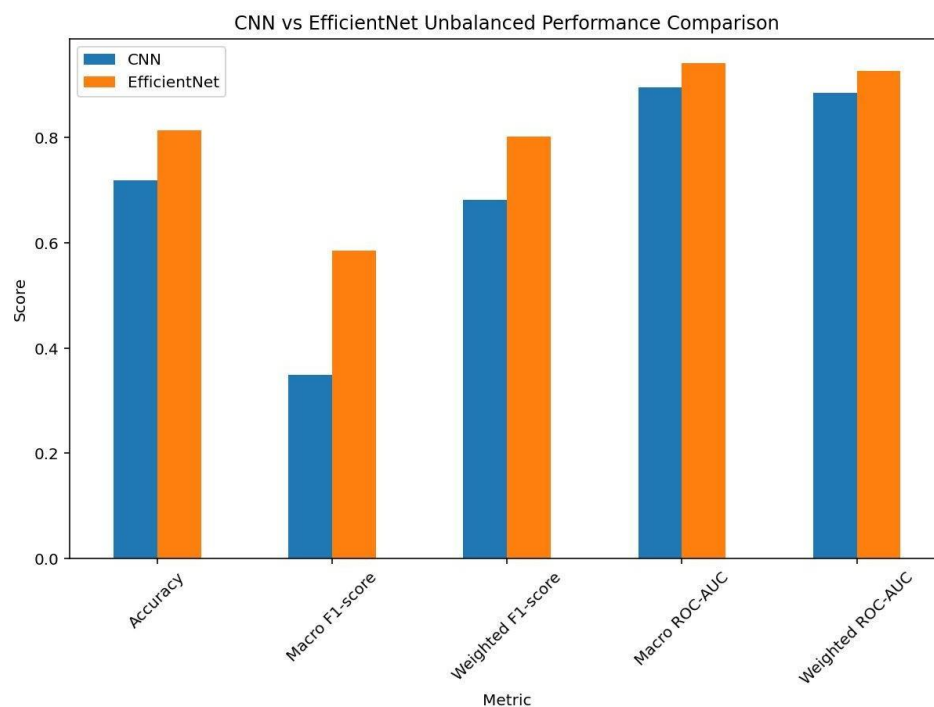


Figure 4.10: Overall performance comparison between CNN and EfficientNet-B0 (unbalanced).

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=== FINAL MODEL COMPARISON ===

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	Metric	CNN	EfficientNet
0	Accuracy	0.7184	0.8133
1	Macro F1-score	0.3489	0.5846
2	Weighted F1-score	0.6817	0.8023
3	Macro ROC-AUC	0.8955	0.9408
4	Weighted ROC-AUC	0.8855	0.9271

Figure 4.11: Final model comparison metrics — CNN vs EfficientNet (unbalanced dataset).

Metric	CNN (Unbalanced)	EfficientNet (Unbalanced)
Accuracy	0.7184	0.8133
Macro F1-score	0.3489	0.5846
Weighted F1-score	0.6817	0.8023
Macro ROC-AUC	0.8955	0.9408
Weighted ROC-AUC	0.8855	0.9271

Table 4.1: Phase 1 final model comparison (unbalanced dataset).

The difference between Macro F1-scores (0.35 vs 0.58) is especially illuminating because although both models managed to achieve a decent accuracy rate because of the prevalence of the majority class, EfficientNet showed significantly better recall rates on average for each class. This can be seen in the ROC graphs for both models, shown in Figures 4.12 and 4.13.

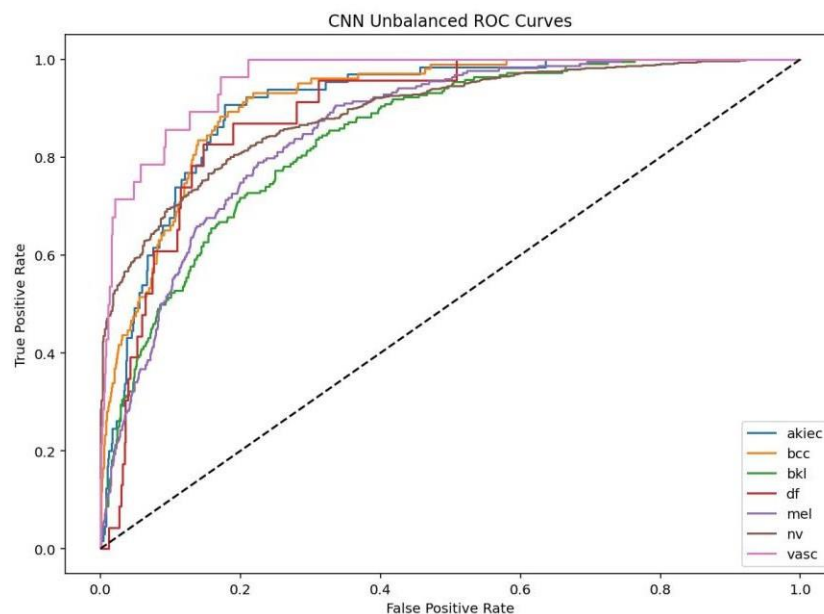


Figure 4.12: ROC curves per class for the CNN (unbalanced dataset).

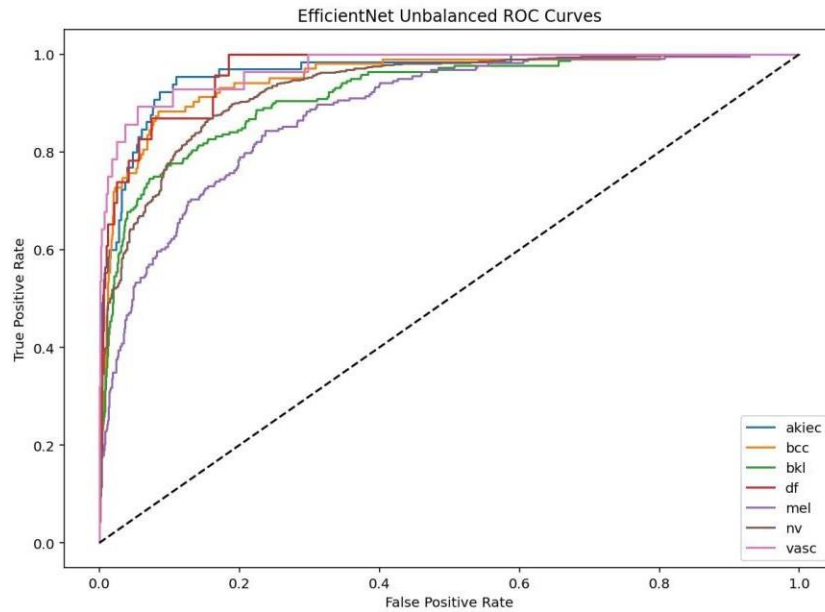


Figure 4.13: ROC curves per class for EfficientNet-B0 (unbalanced dataset).

4.2 Phase 2: Balanced Dataset Experiments

4.2.1 CNN — Balanced Dataset

The custom CNN model that learned from the balanced training data had much better dynamics in terms of learning. After 100 epochs, the values of accuracy for training data and validation data became about 71.4% and 74.0%, respectively (see Figure 4.14). This indicates the effectiveness of the implemented regularization algorithms (batch normalization, dropout), along with the use of balanced training data, in reducing overfitting.

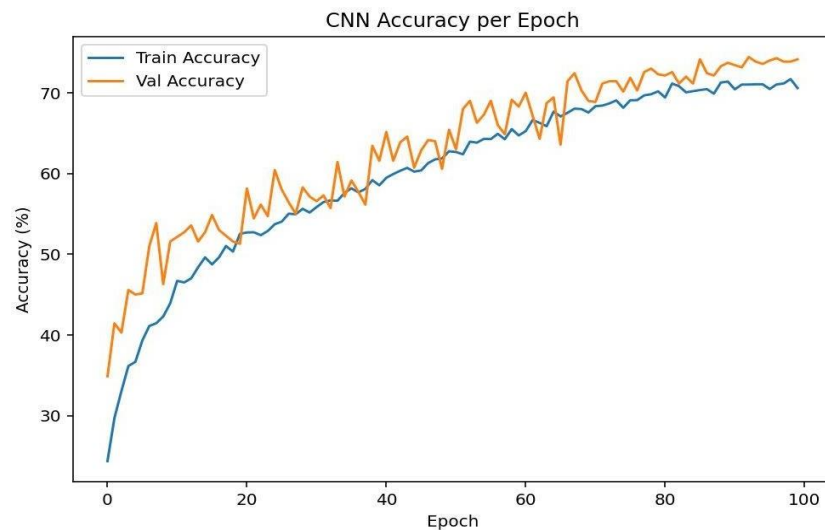


Figure 4.14: CNN (balanced) training and validation accuracy per epoch.

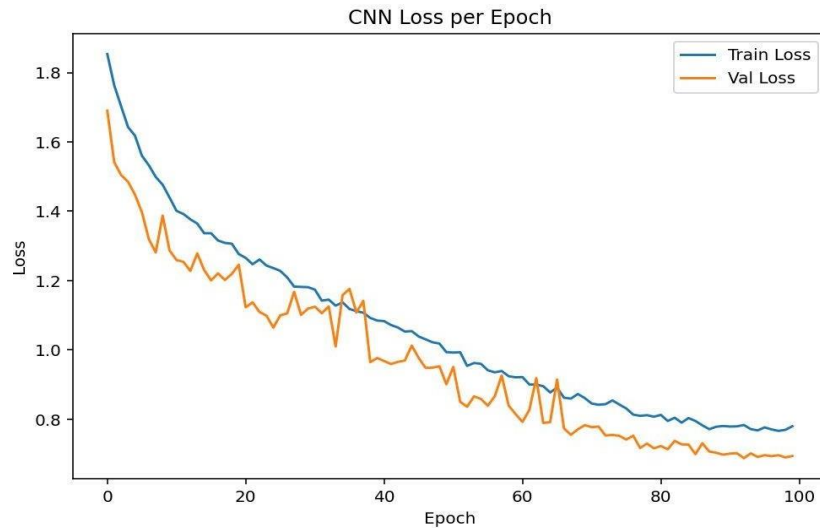


Figure 4.15: CNN (balanced) training and validation loss per epoch.

There is significant improvement in the balance of performances among different classes in the confusion matrix (Fig. 4.16). There were few confusions in NV class, specifically MEL (29), but the high level of biases towards NV witnessed in phase 1 became less prominent. The accuracy of predictions in DF is 93%, while VASC achieved 100% sensitivity. In ROC curve analysis for balanced CNN (Fig. 4.17), most classes exhibited good discrimination capacity, except for NV and BKL.

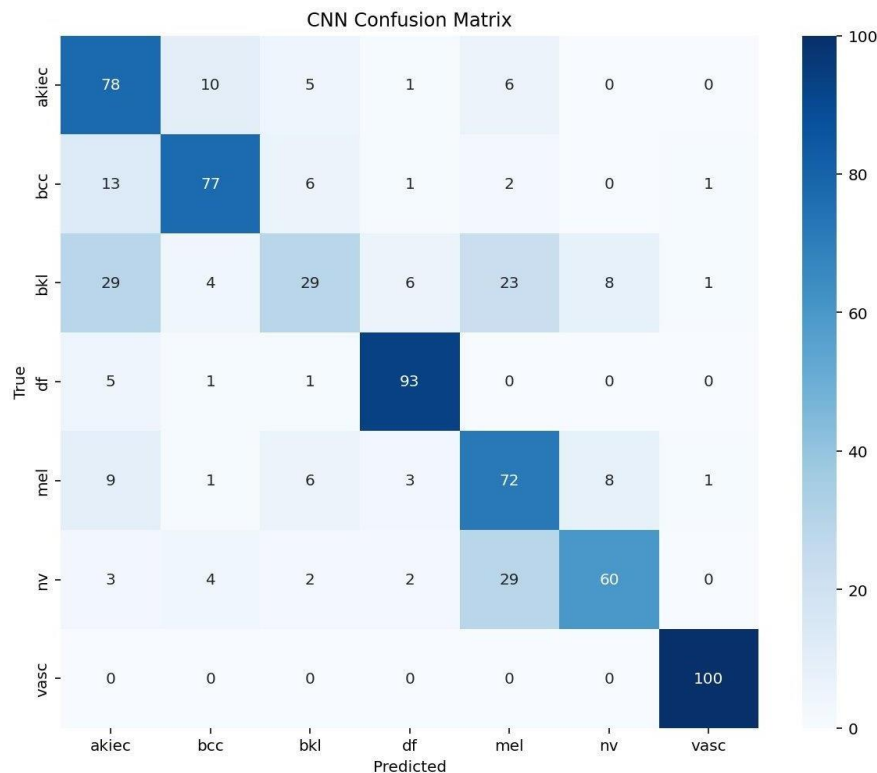


Figure 4.16: Confusion matrix for the CNN trained on the balanced dataset.

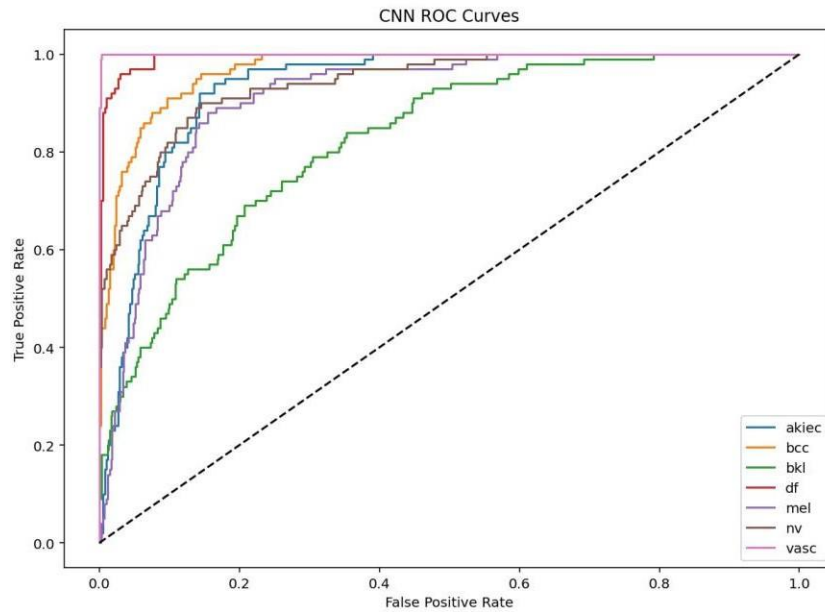


Figure 4.17: ROC curves per class for the CNN (balanced dataset).

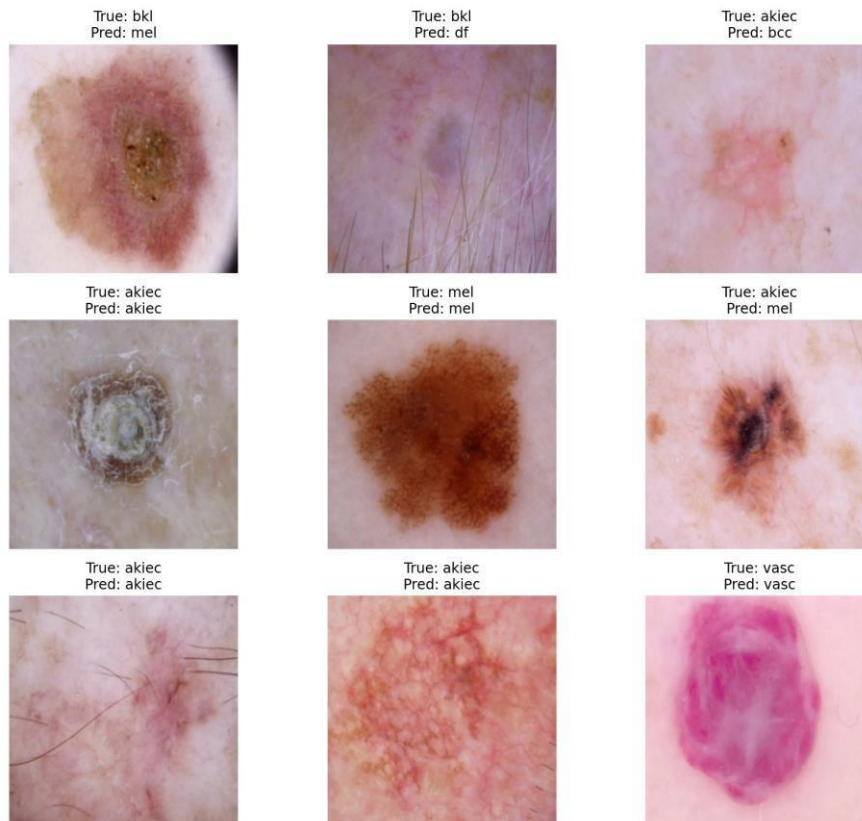


Figure 4.18: Sample predictions from the CNN (balanced dataset).

4.2.2 EfficientNet-B0 — Balanced Dataset

The EfficientNet-B0 model using the balanced data set proved to be a fast learner, with accuracy for training getting close to 92% and validation accuracy close to 86% by epoch 50 (refer Figure 4.19). The loss graphs (refer Figure 4.20) depict an acceptable behavior with the training loss converging to approximately 0.22 and validation loss converging to approximately 0.40, which indicates some amount of overfitting.

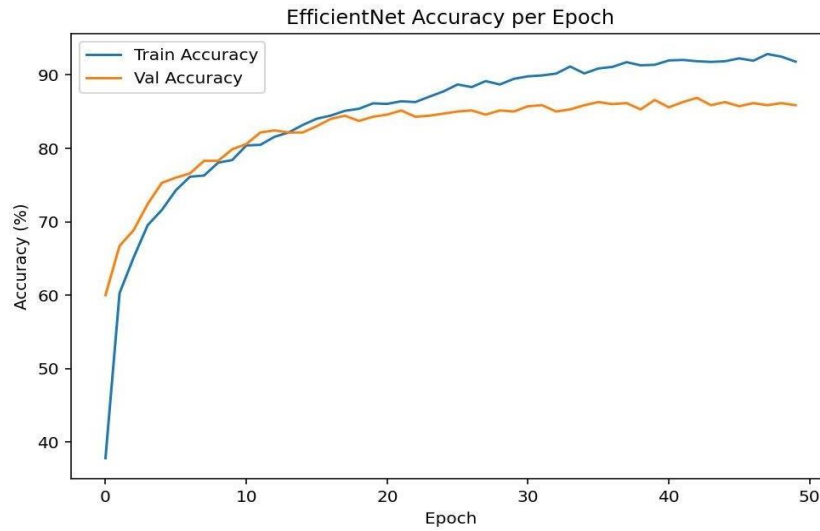


Figure 4.19: EfficientNet-B0 (balanced) training and validation accuracy per epoch.

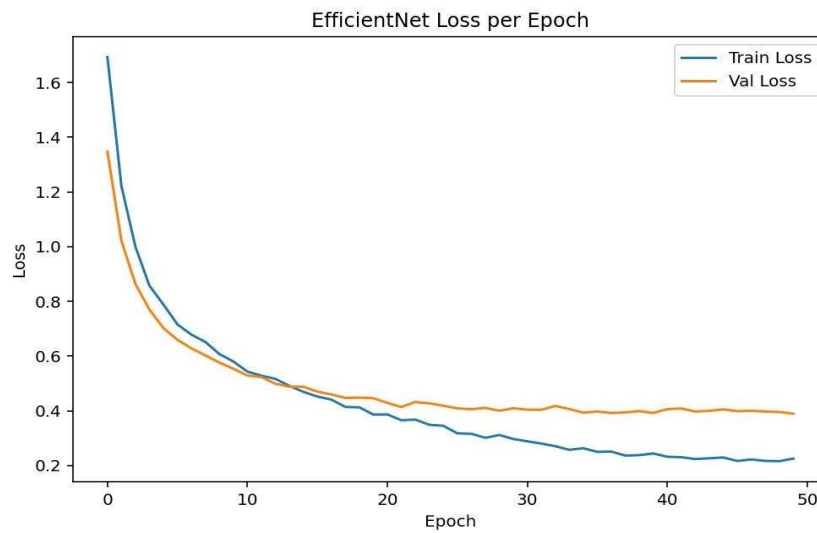


Figure 4.20: EfficientNet-B0 (balanced) training and validation loss per epoch.

From the confusion matrix shown in Figure 4.21, the model performs well in all classes, where DF and VASC have a 100% recall rate, BCC is 96/100, while AKIEC has a 90% rate. The major confusion was in classifying between NV and MEL, which had 18 instances where NV was mistaken for MEL due to their visual similarity. From the ROC curve shown in Figure 4.22, the results lie tightly in the top left corner.

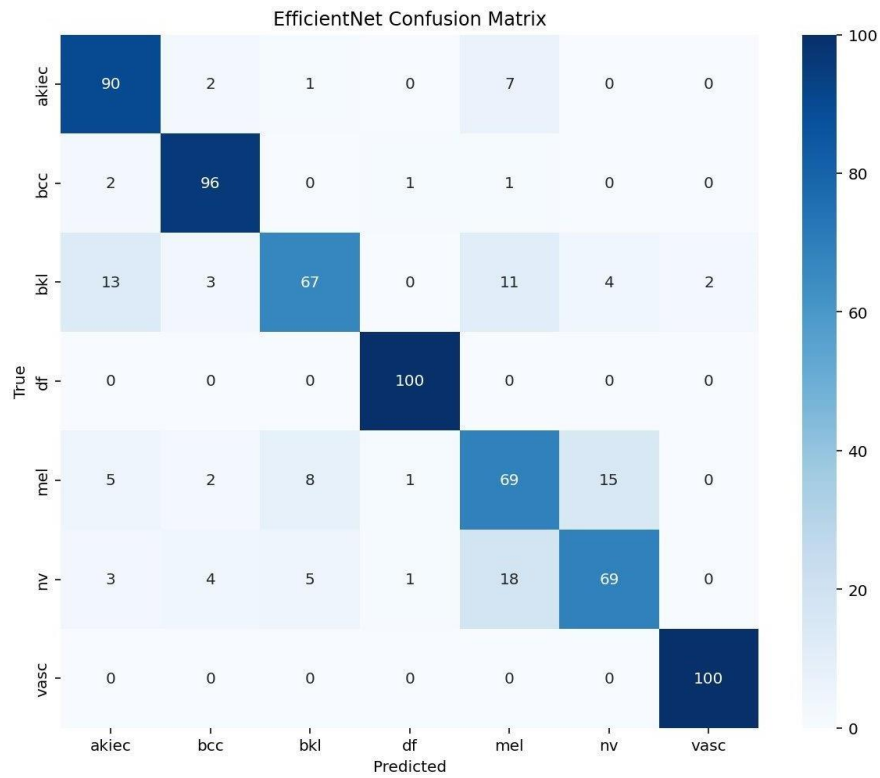


Figure 4.21: Confusion matrix for EfficientNet-B0 (balanced dataset).

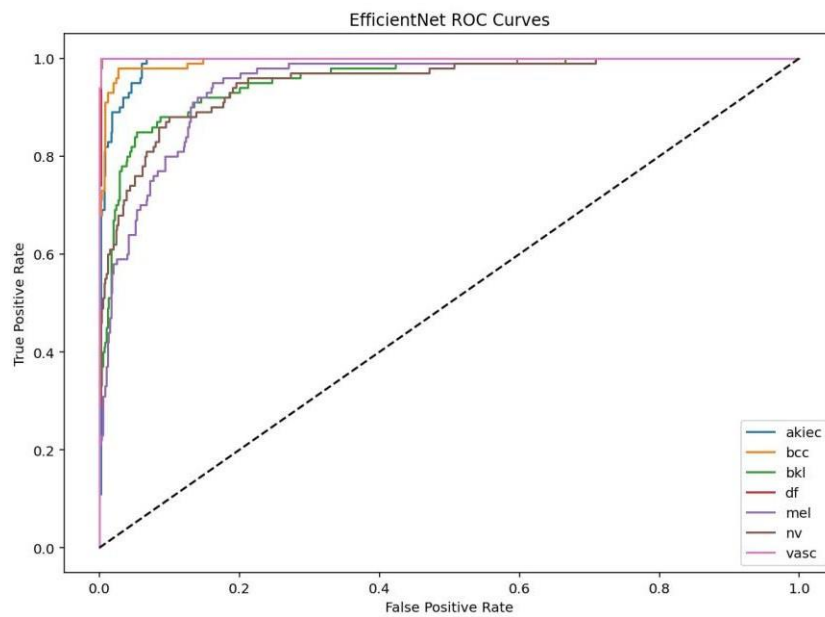


Figure 4.22: ROC curves per class for EfficientNet-B0 (balanced dataset).

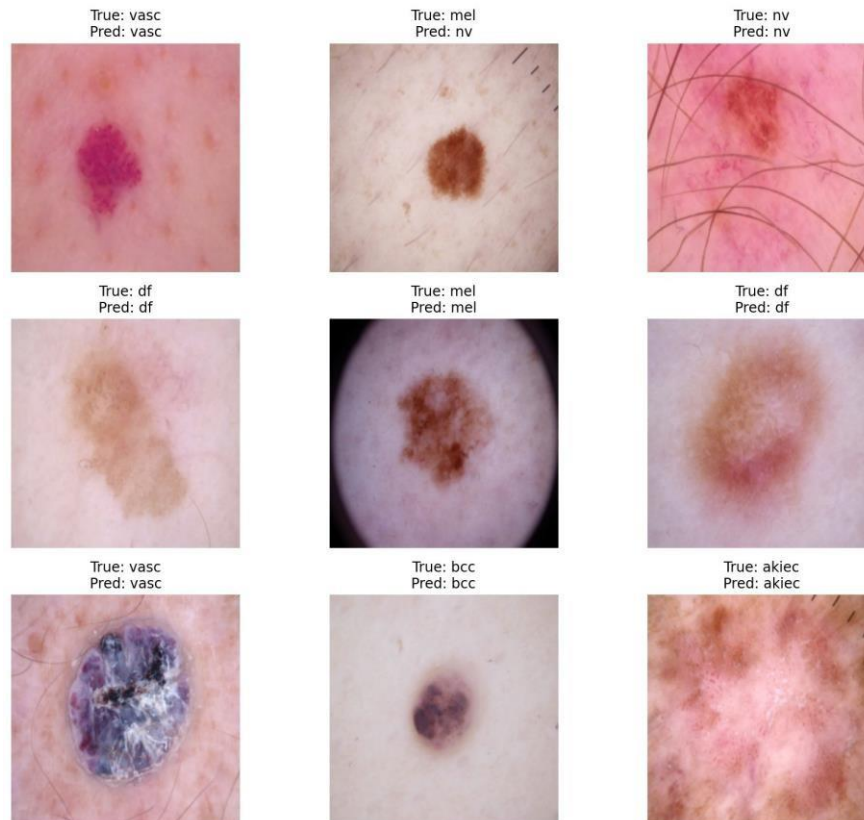


Figure 4.23: Sample predictions from EfficientNet-B0 (balanced dataset).

4.2.3 ResNet50 — Balanced Dataset

The ResNet50 model showed the best performance during the training process compared to other models in Phase 2. The training accuracy attained about 95%, while validation accuracy attained 88% before early stopping at epoch 44 (see Figure 4.24). The loss plots (see Figure 4.25) indicate convergence trends, where validation loss converged at 0.35, which is lower than EfficientNet, with minimal loss difference indicating good generalization ability.

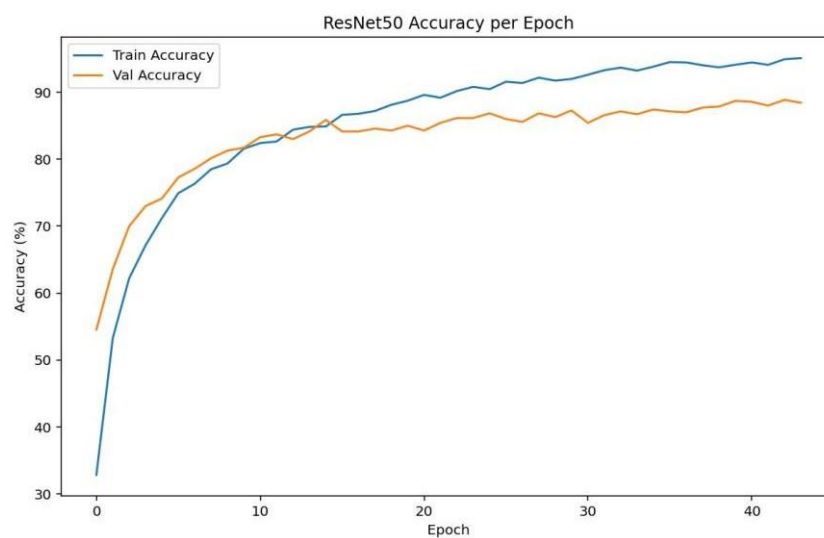


Figure 4.24: ResNet50 (balanced) training and validation accuracy per epoch.

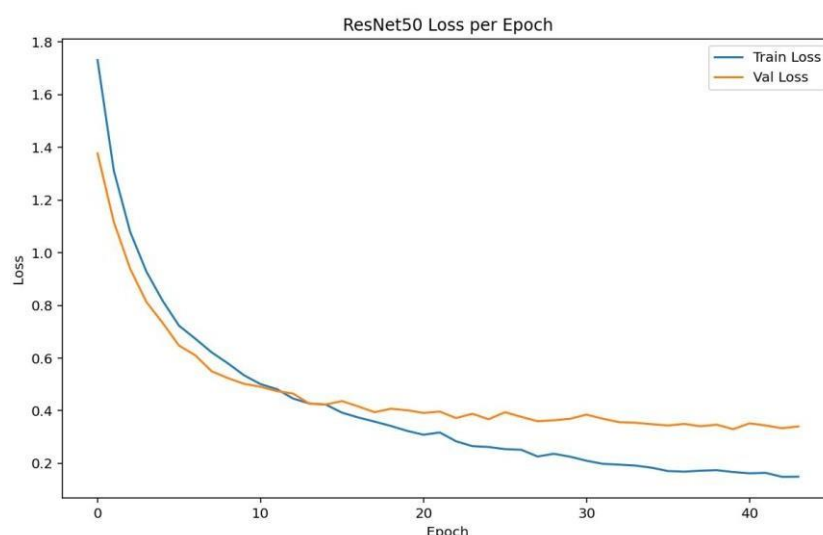


Figure 4.25: ResNet50 (balanced) training and validation loss per epoch.

As observed in the ResNet50 confusion matrix illustrated in Figure 4.26, diagonal elements are relatively larger compared to other models' outputs. For instance, DF and VASC had 100% recall. BCC and AKIEC demonstrated great results with correct predictions of 95 and 92, respectively. Additionally, NV demonstrated 80 correct predictions among 100, where 13 of them were mistaken for MEL, the most frequent confusion pattern across all architectures, attributable to the visual similarity between melanocytic lesions.

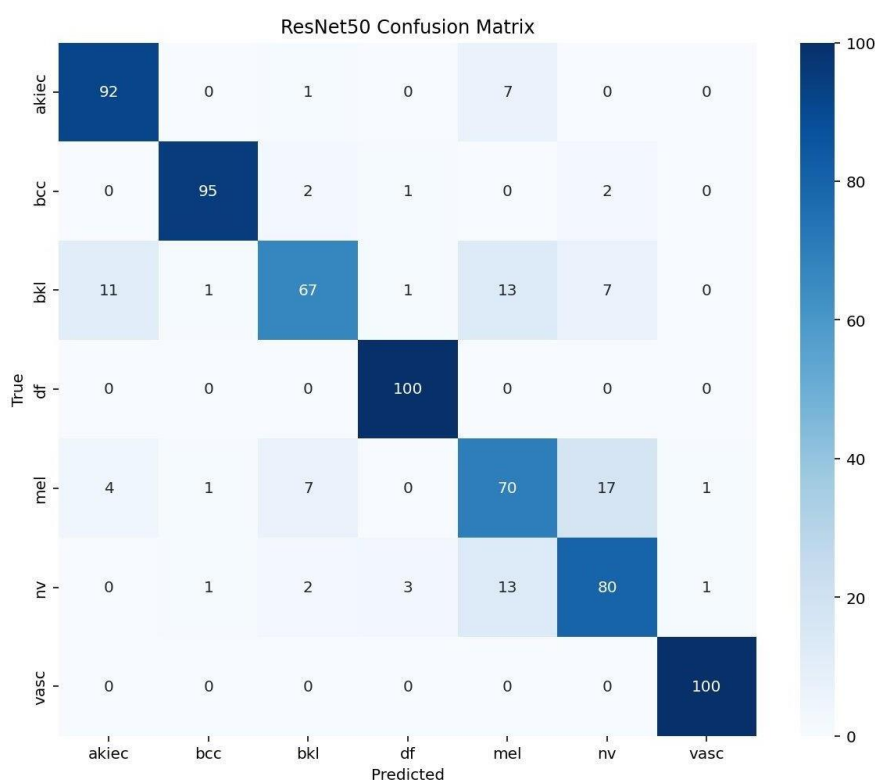


Figure 4.26: Confusion matrix for ResNet50 (balanced dataset).

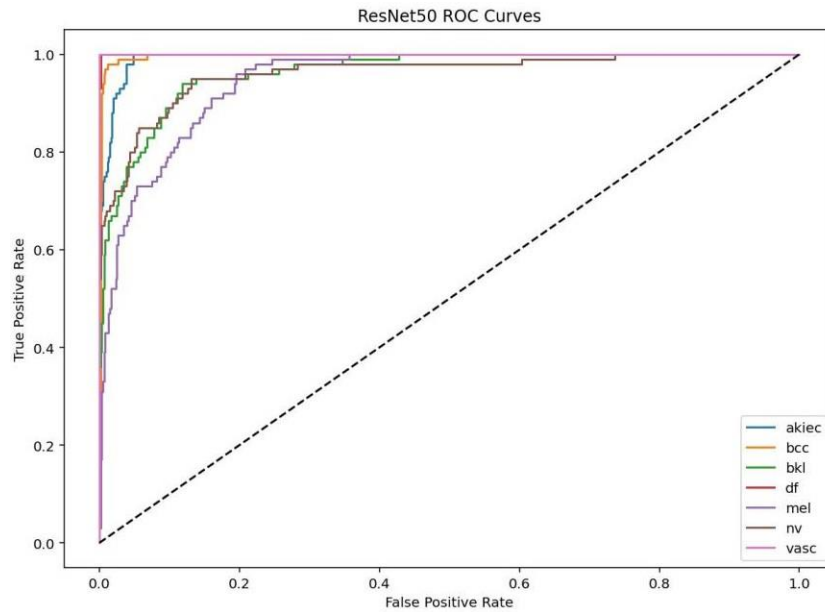


Figure 4.27: ROC curves per class for ResNet50 (balanced dataset).



Figure 4.28: Sample predictions from ResNet50 (balanced dataset).

4.2.4 VGG16 — Balanced Dataset

Convergence was faster in the VGG16 model, which achieved around 89% and 87% training and validation accuracies, respectively, in just 25 epochs before early stopping (Figure 4.29). This is supported by the convergence graph (Figure 4.30), which demonstrates smooth convergence without much fluctuation due to the use of lighter augmentation and low learning rates.

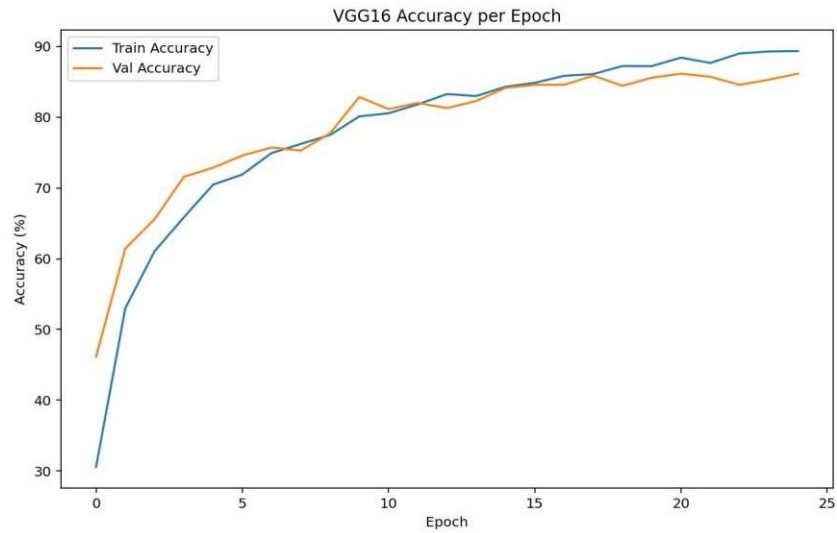


Figure 4.29: VGG16 (balanced) training and validation accuracy per epoch.

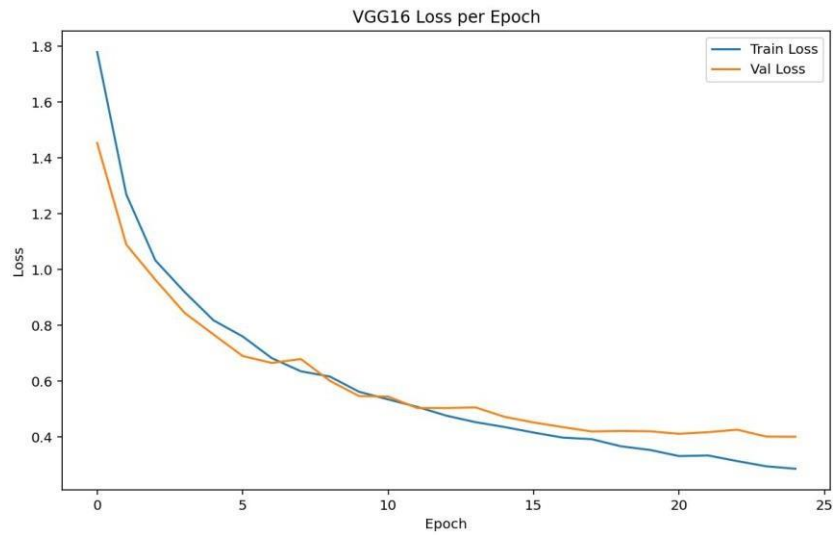


Figure 4.30: VGG16 (balanced) training and validation loss per epoch.

According to the confusion matrix of VGG16 (Fig. 4.31), the performance on DF and VASC datasets is very high (recall rate = 100%) as well as on the BCC and AKIEC classes 87 and 88 correct predictions respectively. The BKL class displayed the highest level of confusion with 12 errors assigned to MEL and 11 to NV, which is probably due to visual similarities between benign keratosis-like lesions and both melanoma and nevi.

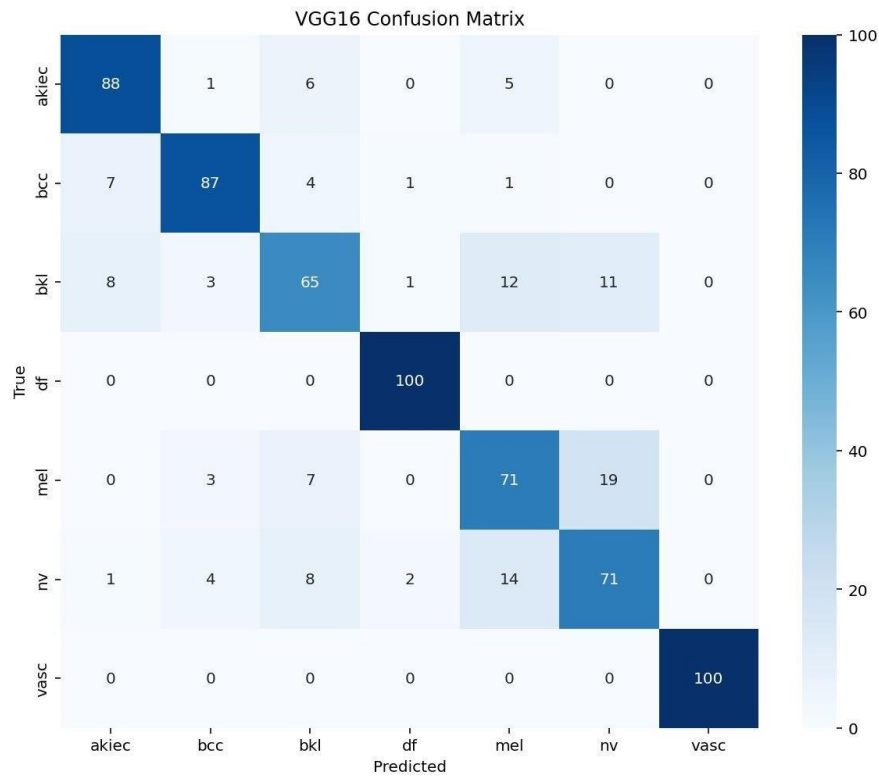


Figure 4.31: Confusion matrix for VGG16 (balanced dataset).

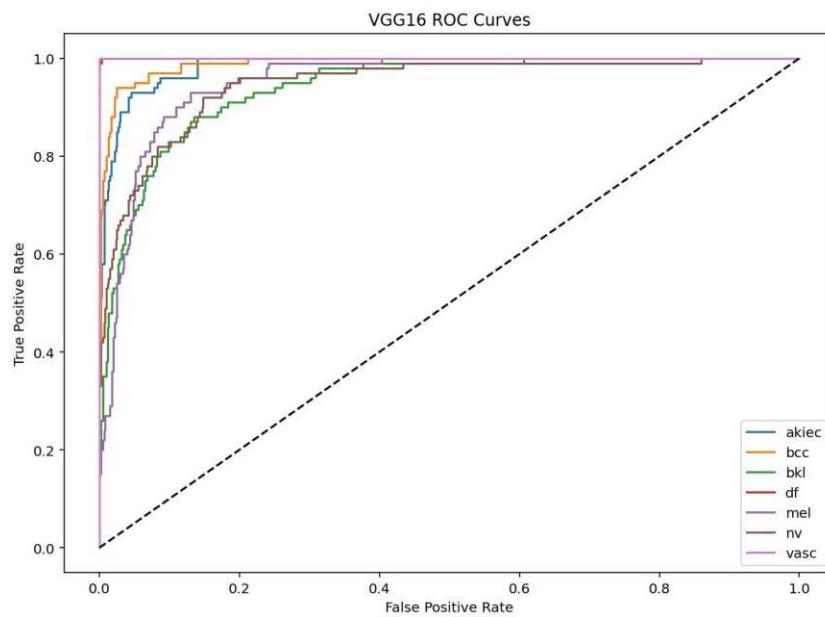


Figure 4.32: ROC curves per class for VGG16 (balanced dataset).



Figure 4.33: Sample predictions from VGG16 (balanced dataset).

4.3 Phase 2 Comparative Analysis

The per-class F1-score comparison between the four Phase 2 models is shown in Figure 4.34. The transfer learning models had significantly better results than the custom CNN model for all the seven classes, with the biggest gaps seen for AKIEC and BKL. However, for the transfer learning models, the differences were not as great the ResNet50 model yielded the best results for NV (0.778) and BCC (0.960), while EfficientNet performed best on DF (0.985).

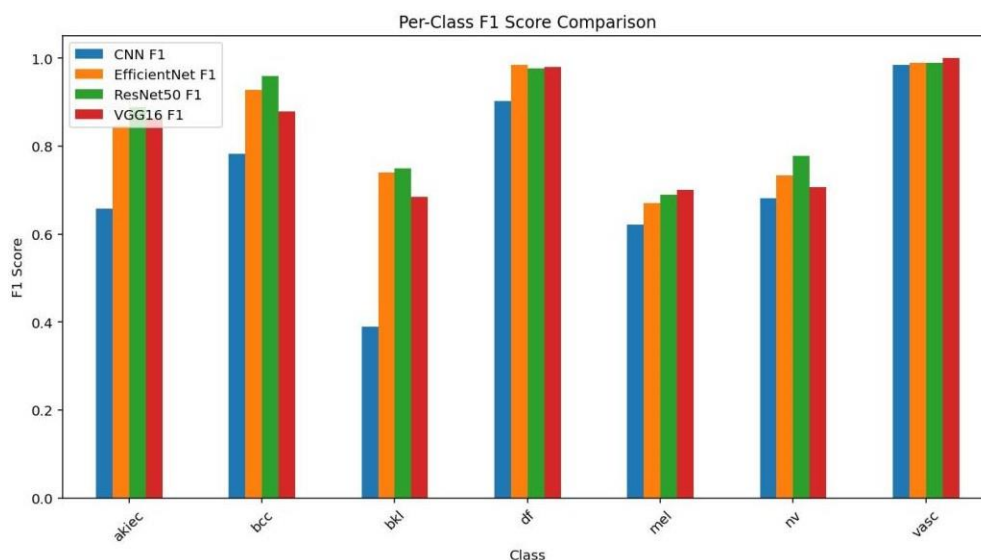


Figure 4.34: Per-class F1-score comparison across all four balanced-dataset models.

```

=== PER-CLASS COMPARISON ===

```

	Class	CNN F1	EfficientNet F1	ResNet50 F1	VGG16 F1
0	akiec	0.658	0.845	0.889	0.863
1	bcc	0.782	0.928	0.960	0.879
2	bkl	0.389	0.740	0.749	0.684
3	df	0.903	0.985	0.976	0.980
4	mel	0.621	0.670	0.690	0.700
5	nv	0.682	0.734	0.777	0.706
6	vasc	0.985	0.990	0.990	1.000

Figure 4.35: Per-class F1-score summary table for all balanced-dataset models.

```

=== OVERALL PERFORMANCE COMPARISON ===

```

	Metric	CNN	EfficientNet	ResNet50	VGG16
0	Accuracy	0.727	0.844	0.863	0.831
1	Macro F1	0.717	0.842	0.861	0.830
2	Weighted F1	0.717	0.842	0.861	0.830

Figure 4.36: Overall performance comparison summary (balanced dataset).

Figure 4.37 presents the overall performance comparison chart, and Figure 4.38 the final model comparison table. Table 4.2 consolidates the key metrics for all Phase 2 models.

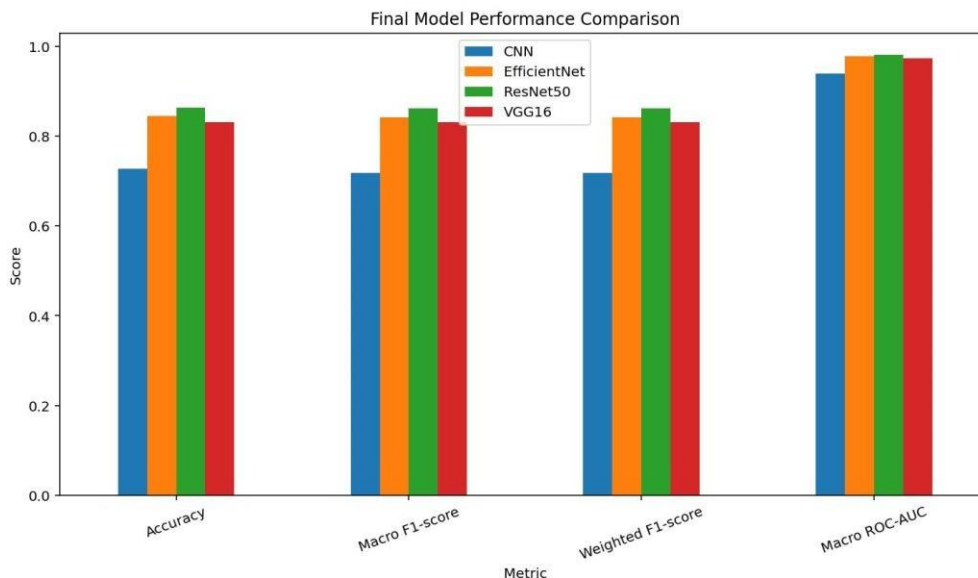


Figure 4.37: Final model performance comparison across all four balanced-dataset architectures.

```

=== FINAL MODEL COMPARISON ===

```

	Metric	CNN	EfficientNet	ResNet50	VGG16
0	Accuracy	0.7271	0.8443	0.8629	0.8314
1	Macro F1-score	0.7171	0.8417	0.8613	0.8303
2	Weighted F1-score	0.7171	0.8417	0.8613	0.8303
3	Macro ROC-AUC	0.9395	0.9771	0.9805	0.9737

Figure 4.38: Final model comparison metrics table (balanced dataset).

Metric	CNN	EfficientNet	ResNet50	VGG16
Accuracy	0.7271	0.8443	0.8629	0.8314
Macro F1-score	0.7171	0.8417	0.8613	0.8303
Weighted F1-score	0.7171	0.8417	0.8613	0.8303
Macro ROC-AUC	0.9395	0.9771	0.9805	0.9737

Table 4.2: Summary of evaluation metrics for all Phase 2 models (balanced dataset).

The final ranking of the models by their combined score is shown in Figure 4.39. ResNet50 came first, scoring 0.8738, followed by EfficientNet (0.8562), VGG16 (0.8451), and CNN (0.7429). The macro-ROC-AUC scores are displayed in Figure 4.40

```
=====
FINAL MODEL RANKING (BEST → WORST)
=====
```

	Model	Accuracy	Macro_F1	Weighted_F1	ROC_AUC	Final Score
1	ResNet50	0.862857	0.861307	0.861307	0.980548	0.873774
2	EfficientNet	0.844286	0.841743	0.841743	0.977079	0.856167
3	VGG16	0.831429	0.830302	0.830302	0.973721	0.845038
4	CNN	0.727143	0.717123	0.717123	0.939483	0.742866

Figure 4.39: Final model ranking (best to worst) based on composite score (balanced dataset).

```
=== ROC-AUC EVALUATION ===

CNN ROC-AUC: 0.9395
EfficientNet ROC-AUC: 0.9771
ResNet50 ROC-AUC: 0.9805
VGG16 ROC-AUC: 0.9737
```

Figure 4.40: Macro ROC-AUC values for all balanced-dataset models.

4.4 Evaluation and Discussion

4.4.1 Effect of Dataset Balancing

The shift from Phase 1 (unbalanced) to Phase 2 (balanced) yielded consistent improvements in macro F1-scores for all the architectures tested. For instance, the macro F1-score for the custom CNN increased from 0.349 to 0.717, a difference of about 0.368, indicating that most of the errors associated with the phase one results were due to class imbalance as conflicted to any design deficiency within the architecture. In the case of EfficientNet, there was an improvement in the macro F1-score from 0.585 to 0.842. More importantly, the increase in performance was observed in minority classes, specifically DF, VASC, and AKIEC.

4.4.2 ResNet50 as the Best-Performing Architecture

The ResNet50 model scored the best overall score because of the higher accuracy (86.3%), F1-score (0.861), and ROC-AUC (0.9805) scores. The ability of the residual connections in the ResNet50 model to pass gradients along skip connections during

the backward pass prevents vanishing gradients which can affect the process of training deep networks. In addition to this feature, the combination of the deep classifier head that makes use of batch normalisation and two dropout layers enabled the network to learn highly discriminative features from the dataset while still being able to generalise well.

4.4.3 Persistent Confusion Between NV and MEL

For all models and both phases, the most common form of misclassification emerged from the relationship between melanocytic nevi (NV) and melanoma (MEL). In the confusion matrix of the ResNet50, 17 cases of MEL were misclassified as NV while 13 NVs were misclassified as MEL. It is true that clinically, there are reports where there exists an overlap between the clinical findings of MEL and NVs since both have irregular pigmentation and color variations making the distinction very difficult for even the most expert clinicians. The same was seen with BKL, which had an overlap with MEL and NV.

4.4.4 Debugging and Evaluation Pipeline Integrity

One of the challenges encountered while comparing the models on Phase 2 of the project was that while evaluating the CNN using the test images, the predictions were mistakenly made using the EfficientNet model due to the variable overwrite problem (where the variable "model" was assigned another value as the EfficientNet model after training the CNN model). Initially, the results obtained had equal metrics for both models; however, this was later rectified by saving each model into a uniquely named variable (cnn_model, efficientnet_model, resnet_model, vgg_model).

4.4.5 Comparison with Literature

The Phase 2 ResNet50 model performs well, reaching an accuracy of 86.3% and a macro-ROC-AUC of 0.9805, which is on par with results reported in the literature. For example, Nguyen et al. (2025) found ResNet50 achieving an 87.51% accuracy rate and VGG19 achieving an 89.68% accuracy rate on an augmented HAM10000 data set. Overall, the results in this study align with other benchmarks in the field demonstrate that the use of ResNet50 with selective fine-tuning and a balanced data set can produce highly reliable results. Finally, the EfficientNet-B0 model's accuracy of 84.4% is comparable to the 84.36% achieved by Alenezi et al. (2024), but again, the performance is significantly higher when the balanced data set is used.

5. Conclusions and Future Work

In this study, automatic multiclass classification of skin lesions using HAM10000 dermoscopic images based on deep learning was studied in two phases, an unbalanced phase and a balanced phase post hybrid resampling. This study achieved all six objectives discussed in Chapter 1.

Objective 1 (Custom CNN Model, unbalanced): The custom CNN scored an accuracy of 71.8%, but a macro F1 score of just 0.349, demonstrating that high accuracy is a deceptive metric when dealing with a highly imbalanced dataset. Majority bias towards NV class resulted in low minority recall.

Objective 2 (Unbalanced EfficientNet-B0): The implementation of transfer learning for ImageNet pre-trained EfficientNet-B0 model achieved an accuracy score of 81.3%, with the macro F1-score reaching up to 0.585, increasing by 0.236 compared to the CNN model.

Objective 3 (resampling approach): Down sampling majority classes and augmenting minority classes until all classes have a target size of 1,000 images resulted in improved performance of the CNN. The increase in the CNN's macro-F1 score after balancing was 0.368, showing that class imbalance and not lack of capacity was the main challenge in Phase 1.

Objective 4 (comparison with four models): The four balanced models were superior to their unbalanced models. ResNet50 was the best model in comparison with 86.3% accuracy, 0.861 macro F1 score, and 0.9805 macro-ROC-AUC value. DF and VASC had almost perfect recall for all transfer learning models, thus validating that the resampling technique addressed the issue of minority class under-representation.

Objective 5 (Model Ranking): The model ranking from the four models is ResNet50 (0.8738) > EfficientNet-B0 (0.8562) > VGG16 (0.8451) > CNN (0.7429). It can be concluded that the best model for the task is ResNet50 due to the residual connection of its layers and the fine tuning of its last layer.

Objective 6 (per-class classification): In terms of misclassifications, the NV–MEL pair was the most recurrent type, due to the similar dermoscopic characteristics of melanocytic lesions. This is well-documented in clinical literature, and therefore, this is the principal aspect that requires improvement in future research.

In conclusion, it is quantitatively shown that the effect of balancing data is much greater on the minority class than the effect of increasing architecture complexity, and that transfer learning along with selective fine-tuning is necessary to ensure consistent performance in this application. The ResNet50 model with hybrid resampling balanced data and multiple layers in the classification head is the best configuration, attaining 86.3% accuracy and a macro-ROC-AUC score of 0.9805.

5.1 Recommendations and Future Work

The following guidelines are suggested for future research. It is anticipated that ensemble learning based on ResNet50, EfficientNet-B0, and VGG16 models will result in improved accuracy, considering the literature reports an accuracy rate of up to 86% in balanced HAM10000 data using ensemble techniques. GAN-based or

diffusion model-based synthetic oversampling can produce more accurate samples of minority classes than the oversampling technique used in this experiment, which may enhance the recall for difficult-to-capture classes like MEL. Vision transformer and Swin transformer are obvious choices for future comparisons, as they perform comparably well with other CNNs on medical imaging tasks. The incorporation of spatial or channel attention (such as CBAM) within the CNN architecture will minimise the issue of NV-MEL confusion by prioritising important lesion areas. Testing on a different dataset, such as ISIC 2019 or 2020, will evaluate how well the model generalises to new distributions. Finally, deploying the model as a clinical prototype will enable testing for efficiency and usability.

6. Project Management Review

This chapter will critically analyze the management of the project from its conception until its completion, using the planning documents and Gantt Charts as well as the progress made during the project. It will examine the success or failure of the project manager during the entire project process.

6.1 Project Management Approach

For this project, the methodology that was chosen is the phase-by-phase iterative process. The whole process was organized and separated into different phases: data set analysis, experiment with an unbalanced data set, data balancing, experiment with a balanced data set, testing, and report writing. Every new phase was based on the outcome of the previous phase. Milestones were self-set weekly to make sure that the project was constantly moving forward, and meetings with the supervisor were organized to ensure the correctness of the process and overcome any difficulties arising in the process.

6.2 Gantt Chart Comparison

6.2.1 Version 1 -- Initial Plan

The first Gantt Chart was drawn up at the beginning of the project definition process, outlining an ideal timeline for the entire duration from project conception until its completion. As per this schedule, planned to spend 3 weeks on the exploration and pre-processing of data set, three weeks on the experiment for unbalanced phase using CNN and EfficientNet, six weeks on balanced phase for all four neural network models, two weeks on result analysis and visualization, and finally three weeks on preparing our report. An image of Version 1 of the schedule is attached in Appendix C.

6.2.2 Version 2 -- First Revision

The first revision was needed about four weeks into the experiment, as it was noticed that the balanced phase experiments took more time than expected. The implementation of the hybrid resampling pipeline, testing of four different architectures of the models using different fine-tuning settings, and debugging the issue regarding variable overwriting in the evaluation pipeline took much longer than planned. Therefore, the number of weeks dedicated to the balanced phase experiments increased to one more week while results analysis decreased from two

weeks to one week. The report writing decreased to one week. Version 2 can be found in Appendix C.

6.2.3 Version 3 -- Final Actuals

The final Gantt Chart reflects the actual timeline. The most significant deviation from both prior plans was the extended debugging period required to resolve the CNN evaluation pipeline error documented in Section 4.4.4, which consumed approximately four additional days beyond the allocation in Version 2. The report writing phase was consequently compressed at the start, though the submission deadline was met. Table 6.1 summarises planned versus actual durations for each project phase.

Phase	Planned (V1)	Revised (V2)	Actual
Dataset exploration and preprocessing	6 weeks	7 weeks	7 weeks
Unbalanced experiments (CNN + EfficientNet)	3 weeks	3 weeks	3 weeks
Data balancing implementation	1 week	2 weeks	2.5 weeks
Balanced experiments (4 architectures)	4 weeks	6 weeks	7 weeks
Evaluation, debugging, and visualisation	2 weeks	2 weeks	2.5 weeks
Report writing	3 weeks	1 weeks	1 week
Total	19 weeks	21 weeks	23 weeks

Table 6.1: Planned vs. actual project phase durations across all three Gantt Chart versions.

6.3 Explanation of Plan Changes

The main reason for the change in the first revision was the failure to anticipate the level of difficulty associated with implementing the balanced phase experiments. It was initially assumed that the adjustments to be made when modifying the CNN and EfficientNet configurations from Phase 1 were straightforward; however, it turned out that each network had unique challenges in terms of debugging of checkpoint loading, optimiser adaptation, and scheduling.

The second modification stemmed from the problem identified in Section 4.4.4. Fixing the bug involved a thorough analysis of around 600 lines of training code related to the variable assignments. Looking back, the problem would never have occurred if the proper variable naming strategy had been employed from the very beginning.

6.4 Reflection on Project Management Performance

6.4.1 Strengths

There were several elements of project management carried out efficiently during the process:

- Time management: The project was completed within the required timeframe despite making changes to the plan twice. Instead of limiting the amount of experimentation, the workload of the first part of report writing was compressed. Consequently, the technical content became well-substantiated.
- Work quality: The use of checkpoints, stratification of data splitting, and separation of model variables after fixing the bug contributed to the high reliability and reproducibility of results. All models were evaluated using an agreed-upon procedure, ensuring a fair comparison.
- Monthly supervisor meetings: Meetings were conducted regularly and utilized effectively. Supervisor's questions about choosing specific approaches to fine-tuning each architecture and selecting the metrics of model performance were discussed successfully.
- Documentation: Source code was thoroughly commented line-by-line with descriptions of significant steps that came in handy when returning to previous configurations during the evaluation phase and writing a methodology chapter.

6.4.2 Weaknesses

Several areas in which the performance of the project could have been enhanced included:

- Risk Management: There was no documented risk register for this project. The overwriting of variable bug as well as the underestimated complexity of balanced-phase was not considered among the potential risks and therefore had no planned mitigation steps. Risk register is given in appendix A.
- Scope Control: Extending the balanced phase to include four architectures rather than two as was originally proposed added to the level of analysis while creating scheduling constraints. A formal scope change process could have led to better planning.
- Verification and Testing: It would have been possible to discover the evaluation pipeline bug earlier through the implementation of assertion checks that would prove that the output from each architecture is different before calculating results.
- Communication about delays: Where a revised plan was needed, it was communicated during the next supervisor's meeting, rather than upfront, at the time of discovery. Timely communication would have enabled earlier guidance on either limiting scope or modifying the schedule.

The project met its technical objectives within the scheduled time frame and had results that are deep and good enough to allow drawing the conclusions that have been made. The difficulties related to management faced during the process did not affect the submission process, but they indicated some areas where better planning would have made things easier.

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Appendices

Appendix A -- Risk Register

Table A.1 presents the risk register for this project. Risks are categorised by likelihood (L: Low, M: Medium, H: High) and impact (L, M, H).

ID	Risk Description	L'hood	Impact	Level	Mitigation Strategy
R01	GPU unavailable or insufficient VRAM	M	H	H	Use Colab/Kaggle as backup; reduce batch size if needed.
R02	Class imbalance causing minority-class bias	H	H	H	Apply hybrid resampling: downsample majority, upsample minority with augmentation.
R03	Model variable overwriting corrupting evaluation results	M	H	H	Store each model in a uniquely named variable; verify metrics differ before recording.
R04	Checkpoint loading failure due to architecture mismatch	M	M	M	Test save/load on a small run first; handle incompatibilities with try/except.
R05	Underestimation of experimentation time	H	M	M	Build buffer weeks into plan; communicate delays to supervisor promptly.
R06	Augmentation introducing unrealistic lesion artefacts	L	M	L	Restrict to rotation-invariant, colour-preserving transforms; inspect samples visually.
R07	Plagiarism or academic integrity issue	L	H	M	Ensure original text; cite all sources in Harvard style; submit through Turnitin.

Table A.1: Project Risk Register.

Appendix B -- Ethics Checklist

The following checklist documents the ethical considerations relevant to this project.

Q: Does the project involve human participants or personal data?

A: No. The HAM10000 dataset is publicly available and fully anonymised. No personal identifiers are present and no new data collection involving human participants was conducted.

Q: Does the project involve clinical or medical decision-making?

A: The models developed are research prototypes intended to demonstrate classification capability. They are not deployed in any clinical setting and do not inform actual medical decisions.

Q: Does the project involve processing of sensitive data?

A: The dermoscopic images are anonymised and publicly distributed under a research licence. No sensitive personal data was processed beyond what is available in the public dataset.

Q: Are there potential harms associated with misclassification?

A: The study acknowledges that misclassification -- particularly false negatives for melanoma -- carries clinical risk in a real-world deployment scenario. This motivates the use of macro F1-score and ROC-AUC as primary evaluation metrics.

Q: Is the dataset used under appropriate licence terms?

A: Yes. The HAM10000 dataset is available under the CC BY-NC 4.0 licence, which permits non-commercial research use with attribution. Attribution is provided in the References section.

Q: Has the work been conducted honestly and transparently?

A: Yes. All results are based on actual model outputs. The evaluation pipeline bug is documented transparently in Section 4.4.4.

Appendix C -- Gantt Charts

Three versions of the project Gantt Chart are presented in this appendix, corresponding to the initial plan (Version 1), the first revision (Version 2), and the final actuals (Version 3). These charts support the discussion in Chapter 6 of this report. Versions 1 and 2 should be inserted as scanned or photographed copies of the original planning documents. Table C.1 presents the final actuals (Version 3) in tabular form.

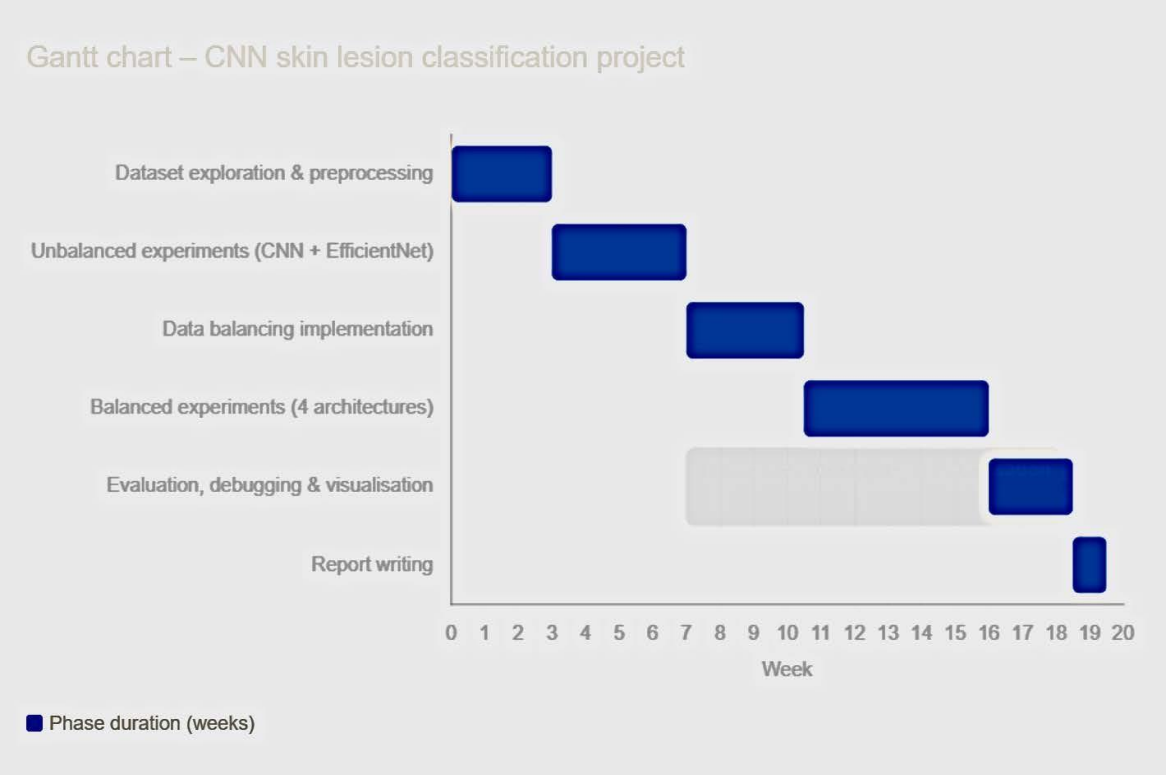


Figure 41: Gantt Chart summary -- Version 3 (Actual Timeline). Insert scanned V1 and V2 charts above this table.

